A Project Report on

Intelligent Tweet Analyzer for Disaster Management

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

By

Hetal Mahajan : 21106014

Christina D'Cruz : 22106024

Prabudh Gaikwad : 21106038

Under the Guidance of

Dr. Rahul K Ambekar

Department of Computer Science & Engineering

(Artificial Intelligence & Machine Learning)

A.P. SHAH INSTITUTE OF TECHNOLOGY, THANE

UNIVERSITY OF MUMBAI

2025-2026

A. P. SHAH INSTITUTE OF TECHNOLOGY, THANE

CERTIFICATE

This is to certify that the project entitled "Intelligent Tweet Analyzer For Disaster Management" is a bonafide work of **Hetal Mahajan (21106014), Christina D'Cruz (22106024), Prabudh Gaikwad (21106038)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning**).

Dr. Rahul K Ambekar Guide	Prof. Sayali P. Badhan Project Co-ordinator
Dr. Jaya Gupta Head of Department	Dr. Uttam D. Kolekar Principal

A. P. SHAH INSTITUTE OF TECHNOLOGY, THANE

Project Report Approval for B.E.

This project report entitled ***Intelligent Tweet Analyzer for Disaster Management*** by ***Hetal Mahajan, Christina D'Cruz and Prabudh Gaikwad*** is approved for the degree of ***Bachelor of Engineering*** in ***Computer Science & Engineering (Artificial Intelligence & Machine Learning)***, ***2025-26***.

Examiner Name	Signature
1._____	_____
2._____	_____

Date:

Place:

Declaration

We declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

(Hetal Mahajan - 21106014)

(Christina D'Cruz - 22106024)

(Prabudh Gaikwad - 21106038)

Date:

Abstract

In recent years, social media platforms such as Twitter (now X) have emerged as critical tools for real-time information exchange during disaster situations. This project, Tweet Analysis for Disaster Management, aims to harness the potential of Twitter data to support faster, data-driven decision-making during crises like floods, earthquakes, and pandemics. The system utilizes Natural Language Processing (NLP) and Machine Learning (ML) techniques to automatically analyze tweets, extract relevant information, and categorize them into actionable types such as Help Requests, Rescue Updates, Blood Donations, and General Information. By integrating geolocation tagging, sentiment analysis, and urgency scoring, the proposed framework enhances situational awareness and enables disaster response teams to prioritize resources effectively. The architecture combines real-time data collection, AI-powered tweet classification, and visualization modules to support timely and targeted interventions. The system addresses challenges such as noisy and multilingual tweet data, ensuring scalability and adaptability across diverse disaster scenarios. Overall, this approach bridges the gap between social media intelligence and operational disaster management, improving response efficiency, coordination, and life-saving outcomes. This project contributes to Sustainable Development Goals (SDG) 11: Sustainable Cities and Communities and SDG 9: Industry, Innovation, and Infrastructure by promoting resilient infrastructure and enhancing disaster response through innovative technological solutions.

Keywords- Disaster Management, Twitter Analysis, NLP, Machine Learning, Sentiment Analysis, Real-Time Response, Geolocation, Emergency Classification

Contents

1. Introduction 1
2. Literature Survey 4
3. Limitation of Existing System 12
4. Problem Statement and Objectives 14
4.1 Problem Statement 14
4.2 Objectives 15
4.3 Scope of the Project 15
5. Proposed System 17
5.1 Architecture of NagarSetu 17
5.2 Data Flow Diagrams 18
Data Flow Diagrams: 18
5.3 Use Case Diagram 21
5.4 Sequence Diagram 22
Analysis: 23
5.5 Activity Diagram 24
Step-by-Step Workflow Description 24
Tweet Collection - 24
Feature Extraction – 25
1. Tweet Classification (Core Analysis) – 25
2. Entity Extraction (NER) – 25
3. Store Classified Data – 25
4. Analytics and Visualization – 25
5. Alert Generation – 26
6. Response Action – 26
7. End – 26
Key Components of the Proposed Framework: 26
Analysis: 26
6. Experimental Setup 28
6.1 Software Requirements 28

6.2 Hardware Requirements 28
6.3 Development Environment 29
6.4 ML/NLP Libraries and Frameworks 29
6.5 Implementation 30
7. Result and Discussion 32
7.1 Result 32
7.2 Discussion 33
8. Conclusion and Future Work 36
References 38
Appendix 40
Publication 43
Acknowledgement 51

LIST OF FIGURES

Figure 5.1.1 Architecture System for System 17
Figure 5.2.0.1 Level 0 DFD of System 18
Figure 5.2.2 Level 1 DFD of System 19
Figure 5.2.3 Level 2 DFD of System 20
Figure 6.5.1 Dashboard 30
Figure 6.5.2 Map View 30
Figure 6.5.3 Analytics 31
Figure 6.5.4 Alerts 31

LIST OF TABLES

Table 2.1 Literature Survey of Existing System 5

ABBREVATION

AI	*Artificial Intelligence*
ML	*Machine Learning*
NLP	*Natural Language Processing*
API	*Application Programming Interface*
UI	*User Interface*
UX	*User Experience*
DL	*Deep Learning*
CNN	*Convolutional Neural Network*
RNN	*Recurrent Neural Network*
LSTM	*Long Short-Term Memory*
BiLSTM	*Bidirectional Long Short-Term Memory*
TF-IDF	*Term Frequency–Inverse Document Frequency*
NER	*Named Entity Recognition*
GPS	*Global Positioning System*
GIS	*Geographic Information System*
DB	*Database*
NoSQL	*Not Only SQL*
JSON	*JavaScript Object Notation*
HTTP	*Hypertext Transfer Protocol*
HTTPS	*Hypertext Transfer Protocol Secure*
REST	*Representational State Transfer*
LLM	*Large Language Model*

STT	*Speech-to-Text*
WER	*Word Error Rate*
F1-Score	*Harmonic Mean of Precision and Recall*

Chapter 1

Introduction

1. Introduction

Social media platforms, particularly Twitter (now X), have become vital real-time information channels during disaster situations, offering a dynamic and immediate source of data that traditional methods, such as news reports or official emergency channels, often cannot match in speed or granularity. In crises like hurricanes, earthquakes, floods, or pandemics, individuals use Twitter to share critical updates, report on-the-ground conditions, request urgent assistance (e.g., medical aid, food, or shelter), offer support (e.g., donations or volunteer services), and disseminate situational awareness (e.g., road closures or safe zones). This user-generated content provides a wealth of real-time, geospatially relevant information that can significantly enhance disaster management efforts, enabling faster decision-making and resource allocation to save lives and mitigate impacts.

These tweets are often noisy, containing informal language, slang, emojis, or irrelevant content, which complicates the identification of urgent needs. For instance, distinguishing a genuine help request from a sarcastic comment or a general expression of concern requires sophisticated processing, as noted by Arvandi et al. (2024). Manual analysis of such data is infeasible due to the scale and time sensitivity, leading to delays in recognizing critical information, such as requests for rescue or medical supplies. Additionally, the diversity of tweet content—ranging from emotional outbursts to actionable updates—further hinders the the ability of first responders and non-governmental organizations (NGOs) to prioritize and act on the most pressing needs.

To address these challenges, the proposed solution is to develop an automated system that leverages advanced natural language processing (NLP) and machine learning techniques to classify tweets into actionable categories, such as Help Request, Blood Donation, Rescue Update, Resource Availability, and General Information. This system would employ algorithms like sentiment analysis, topic modeling, and deep learning models (e.g., BiLSTM-CNN, as used in the 2019 study on Hurricanes Harvey and Irma) to filter and categorize tweets with high accuracy. For example, a tweet stating, 2

"Urgent: Need blood type O+ in Chennai hospital" could be classified as a Blood Donation request, while "Rescue team reached Main Street" could be tagged as a Rescue Update. By incorporating geolocation data and sentiment analysis, as demonstrated in the Chennai flood study (2017), the system can prioritize tweets based on urgency and location, enabling targeted responses.

The impact of such a system is profound for disaster management. By automating the classification of tweets, it enables first responders, NGOs, and government agencies to quickly identify and prioritize urgent needs, such as dispatching rescue teams to stranded individuals or directing medical supplies to areas with critical shortages. This enhances coordination, reduces response times, and optimizes resource allocation, ultimately saving lives and improving efficiency. For instance, Shekhar and Setty (2015) showed how sentiment-based visualization of tweets can map distress levels, aiding responders in focusing on high-priority areas. Furthermore, by integrating with existing disaster management frameworks, the system can bridge the gap between social media insights and operational workflows, addressing a key limitation noted by Arvandi et al. (2024). Despite challenges like noisy data, multilingual

content, and computational demands, as highlighted in the literature, this solution offers a scalable, real-time approach to harness Twitter's potential, making it a game-changer for effective disaster response.

Real-Time Information Hub: Social media platforms, particularly Twitter (now X), serve as critical channels during disasters like hurricanes, earthquakes, floods, or pandemics, providing immediate, user-generated updates that outpace traditional sources like news or official reports.

User Contributions: Individuals post about emergencies (e.g., stranded locations), request help (e.g., medical aid, food), offer support (e.g., donations, volunteering), and share updates (e.g., road closures, safe zones), offering valuable on-the-ground insights.

Sustainability Development Goal Alignment

- **SDG 11: Sustainable Cities and Communities**

The system enhances the ability of cities and communities to prepare for, respond to, and recover from disasters.

By analyzing tweets for real-time information on emergencies, it helps authorities coordinate rescue operations and resource allocation efficiently, contributing to disaster-resilient infrastructure and sustainable urban management.

- **SDG 9: Industry, Innovation, and Infrastructure**

The system demonstrates innovative use of emerging technologies like Natural Language Processing (NLP), Machine Learning (ML), and Large Language Models (LLMs) for societal benefit.

It promotes the integration of digital infrastructure in emergency management systems.

Chapter 2

Literature Survey

2. Literature Survey

Recent research in disaster management has increasingly focused on leveraging social media platforms, particularly Twitter (now X), as real-time information sources during emergencies. Studies have demonstrated that user-generated content can provide critical situational awareness, including requests for help, infrastructure damage reports, and resource availability. Advanced Natural Language Processing (NLP) techniques such as sentiment analysis, topic modeling, and deep learning-based classification (e.g., LSTM, CNN, and transformer models like BERT) have been widely applied to extract meaningful insights from this unstructured data. These approaches significantly improve the speed and efficiency of identifying actionable information compared to traditional manual monitoring systems. However, despite these advancements, many systems still struggle with handling high volumes of noisy, ambiguous, and context-dependent tweet data, which affects classification accuracy and reliability.

Furthermore, existing solutions often face limitations in real-time deployment and scalability, especially during large-scale disasters where millions of tweets are generated within short timeframes.

Table 2.1 Literature Survey of Existing System

Year	Title	Domain	Algorithm	Dataset	Gap
2024	Twitter Analysis in Emergency Management: Recent Research and Trends	Disaster Management, Social Media Analysis	Sentiment Analysis, Topic Modeling, Geolocation-based Analysis	Various disasterrelated tweet datasets (e.g., hurricanes, earthquak es) from the past five years	Limited discussion on multilingual tweet analysis and integration with traditional disaster management systems, which restricts applicability in diverse regions and operational settings.
2023	Social Response and Disaster Management: Insights from Twitter Data Assimilation on Hurricane Ian	Disaster Management, NLP, Spatiotempo ral Analysis	Sentiment Analysis, Topic Modeling	Over 20 million tweets from Hurricane Ian (2022)	Limited handling of noisy or irrelevant tweets, which reduces the accuracy of spatiotemporal insights, and no mention of multimodal (e.g., image) analysis.
2019	A Deep Learning Approach for Tweet Classification and Rescue Scheduling for Effective Disaster Management	Disaster Management, Deep Learning	Attention-based BiLSTM, CNN, Feature Engineering	Tweets from Hurricane s Harvey and Irma	Dependence on labeled datasets and high computational requirements, limiting scalability for real-time use in resourceconstrained environments.

2017	Usage and Analysis of Twitter During 2015 Chennai Flood Towards Disaster Management	Disaster Management , Information Propagation	Random Forests, Other Machine Learning Algorithms	Tweets from the 2015 Chennai Flood	Limited focus on non-English (e.g., Tamil) tweets and lack of integration with realworld disaster response systems, reducing practical utility.
2015	Disaster Analysis Through Tweets	Disaster Management , Sentiment Analysis	Sentiment Classification, Visualization Techniques	Tweet datasets from natural disasters (specific disasters not detailed in the abstract)	Lack of focus on non-English tweets and limited generalizabilit y to different disaster types, potentially missing critical insights from diverse populations.
2024	A multi-model approach for disaster-related tweets classification	Disaster Management / NLP	Ensemble / Multi-model ML (e.g., SVM, RF, DL combined)	Disaster tweet datasets (likely CrisisNLP / Twitter API)	Limited generalization across different disaster types; lacks real-time adaptability
2024	Disaster tweets classification using BERT-based CNN model	NLP / Deep Learning	BERT + CNN hybrid model	Twitter disaster datasets (CrisisNLP or Kaggle)	High computational cost; model complexity makes deployment difficult
2020	I-AID: Identifying actionable information from disaster tweets using BERT and graph attention networks	NLP / Crisis Informatics	BERT + Graph Attention Networks (GAT)	Crisis-related Twitter datasets	Requires large labeled datasets; graph construction overhead

2023	CrisisMatch: Semi-supervised few-shot learning for disaster tweet classification	NLP / Few-shot Learning	Semi-supervised + Few-shot Learning	Limited labeled disaster tweet datasets	Performance drops with noisy data; limited scalability
2023	Natural disaster tweets classification using multimodal data	NLP + Computer Vision	Multimodal Deep Learning (Text + Image models)	Multimodal disaster datasets (tweets + images)	Integration complexity; missing data (text/image imbalance)

In 2024, the paper *"Twitter Analysis in Emergency Management: Recent Research and Trends"* examined various methods for analyzing social media data during disasters. This study focused on domains such as disaster management and social media analysis, employing techniques like sentiment analysis, topic modeling, and geolocation-based analysis. The research utilized diverse tweet datasets related to hurricanes and earthquakes collected over the past five years. However, it highlighted a significant gap—the limited discussion of multilingual tweet analysis and integration with traditional disaster management systems, which restricts its applicability across diverse regions and operational contexts. [1]

In 2023, the study *"Social Response and Disaster Management: Insights from Twitter Data Assimilation on Hurricane Ian"* focused on natural language processing (NLP) and spatiotemporal analysis to assess public sentiment and response during Hurricane Ian (2022). Using a dataset of over 20 million tweets, the researchers applied sentiment analysis and topic modeling to derive insights. Despite its robust scale, the study struggled with noisy or irrelevant tweets, which impacted the precision of spatiotemporal insights, and it did not include multimodal data such as images, limiting the richness of disaster analysis. [2]

A 2019 study titled *"A Deep Learning Approach for Tweet Classification and Rescue Scheduling for Effective Disaster Management"* introduced deep learning methods like Attention-based BiLSTM, CNN, and advanced feature engineering for tweet classification and rescue coordination. Using tweets from Hurricanes Harvey and Irma, this work improved tweet classification accuracy. However, it relied heavily on labeled datasets and required high computational resources, posing challenges for real-time use in resource-constrained environments. [3]

In 2017, the paper *"Usage and Analysis of Twitter During 2015 Chennai Flood Towards Disaster Management"* applied Random Forests and other machine learning algorithms to study information propagation on social media during the 2015 Chennai floods. Although it provided useful insights into how Twitter facilitates real-time communication during crises, the research had limited focus on nonEnglish (particularly Tamil) tweets and lacked integration with operational disaster response systems, reducing its real-world utility. [4]

In 2015, the study *"Disaster Analysis Through Tweets"* explored sentiment classification and visualization techniques to interpret social media data during natural disasters. The study utilized tweet datasets from various disasters, although it did not specify which events. While it presented foundational ideas in disaster sentiment analysis, it had key limitations such as ignoring non-English tweets and offering limited generalizability across disaster types, potentially missing valuable insights from diverse populations. [5]

This paper proposes a multi-model approach for classifying disaster-related tweets by combining multiple machine learning and deep learning models to improve classification accuracy. The system leverages ensemble techniques to capture diverse linguistic patterns present in social media data during disasters. By integrating outputs from different models, it enhances prediction robustness compared to single-model approaches.However, the method faces challenges in generalizing across different disaster scenarios and may struggle with real-time processing due to increased computational complexity. [6]

The authors introduce a hybrid deep learning model that combines BERT with a Convolutional Neural Network (CNN) for disaster tweet classification. BERT is used to extract contextual embeddings, while CNN captures local textual features, resulting in improved classification

performance. The model demonstrates strong results on benchmark disaster tweet datasets. Despite its effectiveness, the approach is computationally expensive and may not be suitable for real-time deployment in resource-constrained environments. [7]

This study presents I-AID, a framework that identifies actionable information from disaster-related tweets using BERT and Graph Attention Networks (GAT). The model not only classifies tweets but also extracts meaningful relationships between them through graph structures, enabling better understanding of crisis-related information. The approach enhances situational awareness during disasters but relies heavily on large labeled datasets and involves complex graph construction, which can limit scalability and efficiency. [8]

The CrisisMatch model focuses on semi-supervised and few-shot learning techniques to classify disaster tweets with minimal labeled data. It is designed to perform effectively even when only a small amount of annotated data is available, making it suitable for emerging disaster scenarios. The model improves adaptability and reduces dependency on large datasets. However, its performance may degrade in the presence of noisy or highly unstructured data, and scalability remains a concern. [9]

This paper explores a multimodal approach for classifying natural disaster tweets by combining textual and visual data. The model integrates deep learning techniques to process both tweet text and associated images, leading to more comprehensive analysis and improved classification accuracy. While the multimodal approach enhances performance, it introduces challenges such as increased model complexity, higher computational requirements, and issues related to missing or imbalanced data across modalities. [10]

This paper studies how Twitter is used during emergency events and identifies patterns in information sharing. The authors analyze tweets during natural disasters to understand communication behavior. The study highlights the importance of social media for situational awareness. It also emphasizes challenges in filtering relevant information from large datasets. [11]

The authors explore digital volunteerism during crisis events using Twitter data. They show how online communities help in organizing information and supporting disaster response. The paper demonstrates the role of crowdsourcing in crisis management. However, it also raises concerns about misinformation and data reliability. [12]

This work provides a comprehensive survey of processing social media messages in mass emergencies. It discusses classification, information extraction, and event detection techniques. The paper highlights the need for automated systems for real-time analysis. It also identifies challenges such as scalability and multilingual data. [13]

The study focuses on crisis informatics and analyzes the types of information shared during disasters. It categorizes tweets into useful and non-useful information. The paper emphasizes the importance of filtering relevant data for emergency responders. It also discusses issues related to data quality and noise. [14]

This paper introduces a system for identifying actionable information from disaster-related tweets. It focuses on extracting requests for help and resource needs. The system uses machine learning models to improve classification. It demonstrates the usefulness of Twitter data for decision support systems. [15]

The authors analyze the role of social media in crisis communication and coordination. The study highlights how platforms like Twitter support real-time information sharing. It also discusses the integration of social media into official emergency systems. Limitations include lack of structured data handling. [16]

This paper proposes an earthquake detection system using Twitter as a real-time sensor. It applies machine learning to detect earthquake-related tweets and estimate locations. The system shows high accuracy and rapid response time. It demonstrates the potential of social media for early warning systems. [17]

The study examines the use of Twitter data for understanding disaster events and human behavior. It analyzes spatial and temporal patterns of tweets during crises. The findings support the use of social media for situational awareness. However, data reliability remains a concern. [18]

This research explores the use of Twitter for detecting earthquakes and analyzing public response. It compares social media signals with traditional seismic data. The results show that Twitter can provide faster detection in some cases. However, false positives can occur due to unrelated tweets. [19]

The authors present a real-time crisis mapping system using Twitter data. The system extracts geolocation information and visualizes disaster impact. It helps emergency responders in identifying affected areas quickly. Limitations include incomplete location data in tweets. [20]

Chapter 3

Limitation of Existing System

3. Limitation of Existing System

The use of Twitter (now X) for disaster management has shown great potential, but existing systems face several limitations that hinder their effectiveness. Below is a concise overview of the key limitations, tailored for your project report, based on the reviewed literature and general challenges in the field.

1. **Noisy and Unstructured Data**: Tweets often include slang, emojis, misspellings, or irrelevant content, complicating accurate information extraction. This noise can lead to misclassification of urgent needs, such as mistaking sarcastic posts for genuine distress calls.

2. **Limited Generalizability**: Systems trained on specific disasters (e.g., hurricanes) often perform poorly on others (e.g., pandemics), requiring retraining and limiting their scalability across diverse crisis scenarios.

3. **Language and Data Bias**: Most systems focus on English tweets, excluding non-English-speaking communities, and urban user dominance skews insights, potentially overlooking rural or underrepresented groups.

4. assignment. Voice recognition fares even worse, with standard Google Speech-to-Text **Real-Time Processing Challenges**: Analyzing millions of tweets in real time demands high computational resources, causing delays that can impede timely disaster response.

5. **Lack of Actionability**: Distinguishing actionable tweets (e.g., help requests) from general commentary is difficult, reducing the ability to prioritize critical needs effectively.

Twitter's potential in disaster management is significant, but current systems face limitations. The noisy and unstructured nature of tweet data, including informal language, complicates natural language processing, leading to misclassification of critical information. Limited generalizability across different disaster types and language biases also hinder their utility. Real-time processing constraints and lack of actionability in tweet classification further hinder performance. To improve tweet analysis systems, advancements in noise reduction, multilingual and multimodal analysis, scalable processing, and integration with operational disaster management frameworks are needed.

Chapter 4

Problem Statement and Objectives

4. Problem Statement and Objectives

4.1 Problem Statement

During disaster situations, Twitter (now X) serves as a critical real-time information channel, with users posting updates, help requests, and support offers. However, the high volume, noise, and unstructured nature of tweets make it challenging to quickly identify and prioritize actionable information, such as urgent needs or critical updates, hindering effective disaster response. Existing systems struggle with noisy data, limited generalizability across disaster types, and language biases, often failing to extract precise location data or assess the urgency of tweets accurately. The proposed solution aims to develop an automated system that addresses these challenges by: extracting and tagging locations for affected areas to enable geospatial prioritization; scoring urgency and performing sentiment analysis for distress detection to identify critical needs; and categorizing tweets into actionable emergencies (e.g., Help Request, Blood Donation, Rescue Update) to facilitate rapid and targeted responses by first responders and NGOs. This system seeks to enhance disaster management efficiency, reduce response times, and improve resource allocation to save lives and mitigate impacts.

4.2 Objectives

- To develop an automated system that analyzes real-time Twitter (X) data during disaster situations.
- To extract and identify relevant disaster-related information from large volumes of unstructured tweets.
- To classify tweets into actionable categories such as Help Requests, Rescue Updates, Blood Donations, and General Information.
- To perform sentiment analysis and urgency detection to prioritize critical situations effectively.
- To extract and tag geolocation data from tweets for identifying affected areas and enabling targeted response.
- To reduce noise and irrelevant content in social media data using NLP techniques.
- To assist disaster management teams and NGOs in faster decision-making and resource allocation.
- To improve response time during emergencies by providing real-time alerts and insights.
- To build a scalable and efficient system capable of handling high-volume social media data.
- To enhance coordination between authorities and responders through data-driven insights.

4.3 Scope of the Project

The scope of this project involves developing an automated system to analyze Twitter (now X) data for enhancing disaster management during crises such as hurricanes, floods, earthquakes, and pandemics. The system will focus on: extracting and tagging locations from tweets to map affected areas for targeted response; scoring urgency and conducting sentiment analysis to detect distress and prioritize critical needs; and categorizing tweets into actionable emergencies (e.g., Help Request, Blood Donation, Rescue Update) to facilitate rapid decision-making by first responders and NGOs. The project will leverage natural language processing (NLP) and machine learning techniques, such as BiLSTM-CNN and Random Forests, to process large-scale, real-time tweet data while addressing challenges like noise, multilingual content, and computational demands. The system aims to integrate with existing disaster management frameworks to improve response efficiency, resource allocation, and situational awareness, with potential applications in real-time monitoring, rescue prioritization, and public sentiment tracking across diverse disaster scenarios.

- **Location Extraction and Tagging:** Map affected areas from tweets for targeted rescue and relief efforts.

- **Urgency Scoring and Sentiment Analysis:** Detect distress and prioritize critical needs using real-time analysis.

- **Actionable Tweet Categorization:** Classify tweets into categories like Help Request, Blood Donation, and Rescue Update for rapid response coordination.

Chapter 5

Proposed System

5. Proposed System

5.1 Architecture of Tweet Analyzer

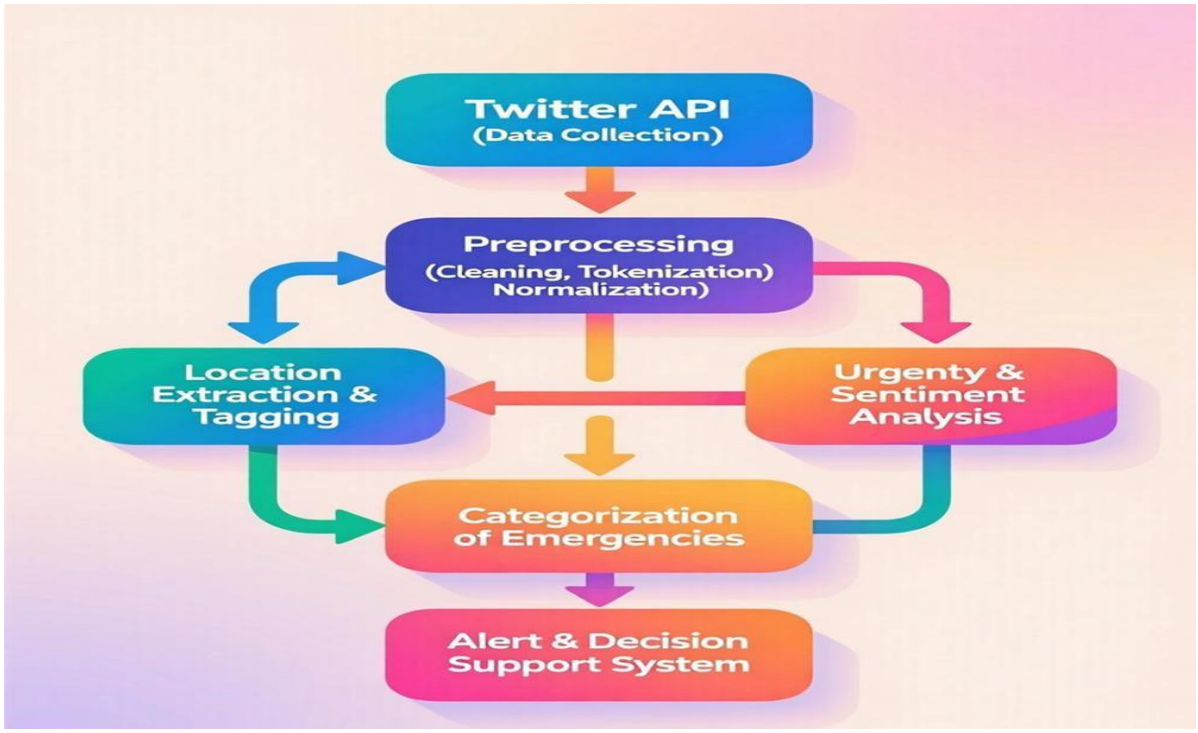

Fig 5.1.1 Architecture of Tweet Analyzer

5.2 Data Flow Diagrams

Data Flow Diagrams:

The Data Flow represents the logical flow of information within the Tweet Analysis for Disaster Management System. It illustrates how raw tweet data moves through various processes from collection to analysis and finally visualization to support timely disaster response and decision- making.

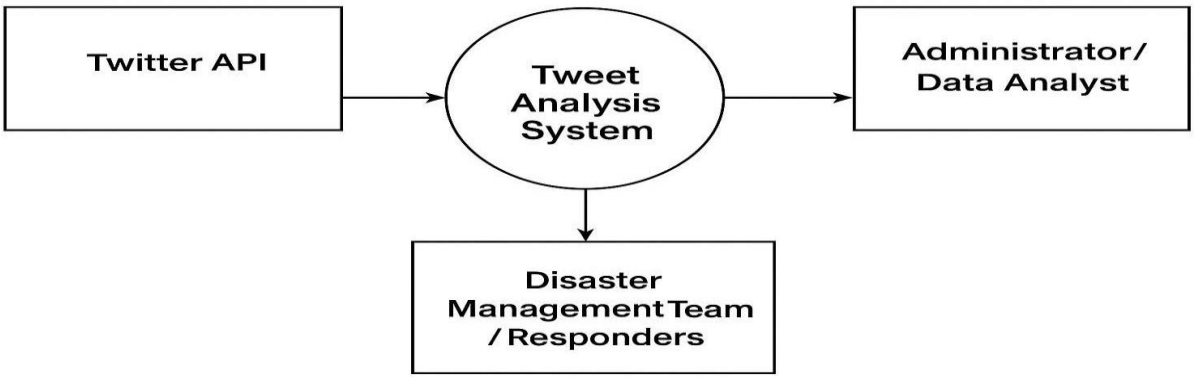

Fig 5.2.0.1 Level 0 DFD of System

At the highest level, the system interacts with three primary external entities:

- Twitter API: Provides real-time disaster-related tweets that serve as the raw data input for the system.
- Administrator / Data Analyst: Manages system configuration, monitors analytics, and validates outputs.
- Disaster Management Team / Responders: Utilize the processed results, such as alerts and situational insights, to coordinate rescue and relief efforts.

The Tweet Analysis System acts as a central processor that:

1. Fetches tweets from the Twitter API based on disaster-related keywords and geolocation filters.
2. Processes and classifies tweets into predefined categories such as Help Request, Rescue Update, Blood Donation, or General Information.
3. Generates visual dashboards and alerts for use by analysts and responders.

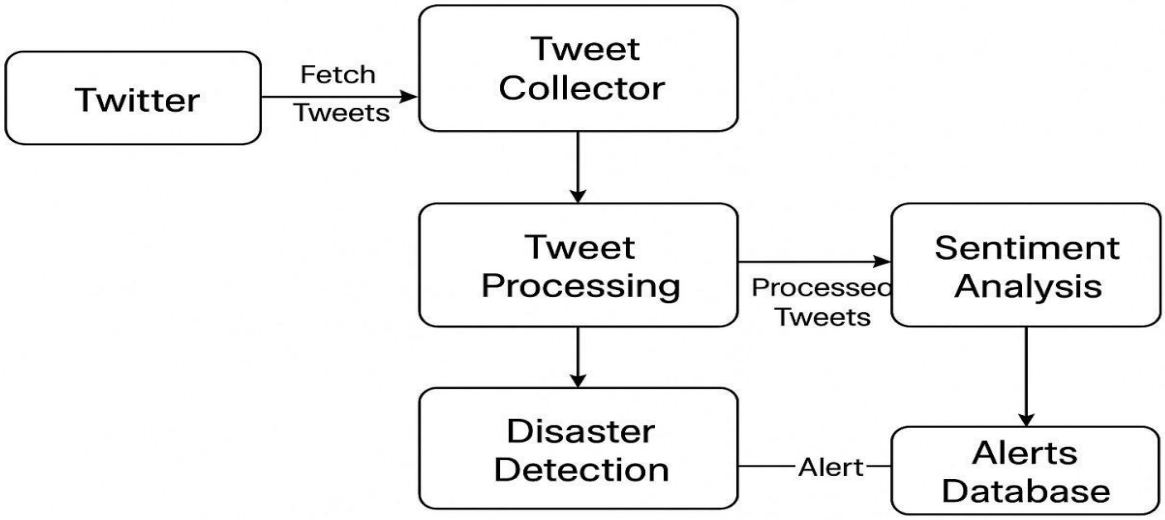

Fig 5.2.2 Level 1 DFD of System

The Level 1 DFD breaks down the internal functions of the system into detailed modules:

1. Process 1.0 – Data Collection Module
    - Input: Tweets from Twitter API.
    - Function: Continuously collects tweets related to disasters using keyword and location- based filters.
    - Output: Raw tweets stored in the Tweet Database.
2. Process 2.0 – Data Preprocessing Module

- Input: Raw tweet data.
- Function: Cleans and normalizes tweets by removing stop words, URLs, special characters, and emojis.
- Output: Cleaned tweets ready for feature extraction.

3. Process 3.0 – Feature Extraction & Classification Module

- Input: Cleaned tweets.
- Function: Converts text into numerical embeddings (TF-IDF or word vectors) and uses NLP models (such as Google Gemini LLM or BiLSTM-CNN) to classify tweets into actionable categories based on urgency or content type.
- Output: Classified tweets stored in the Processed Data Repository.

4. Process 4.0 – Analytics & Visualization Module

- Input: Classified tweet data.
- Function: Aggregates and analyzes results to generate insights such as tweet density per region, resource demand hotspots, and sentiment patterns.
- Output: Interactive dashboards, heatmaps, and analytical reports displayed to the disaster management team.

5. Process 5.0 – Alert Generation Module

- Input: High-urgency tweets and classification results.
- Function: Automatically generates alerts for first responders and NGOs.
- Output: Notifications and alerts sent to the disaster management dashboard or mobile devices

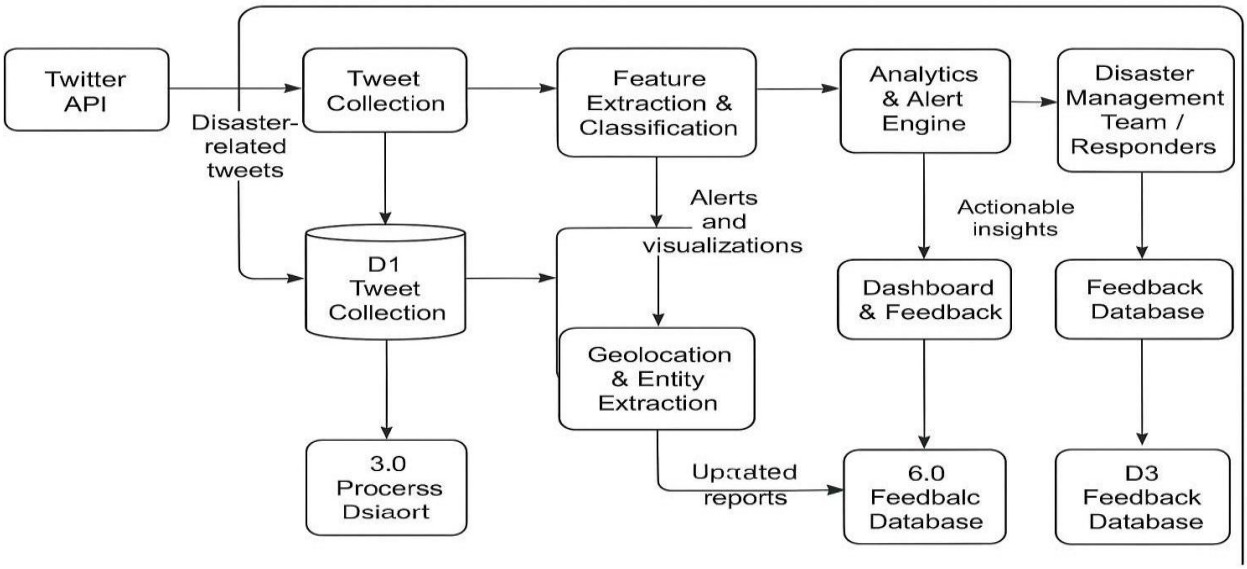

Fig 5.2.3 Level 2 DFD of System

5.3 Use Case Diagram

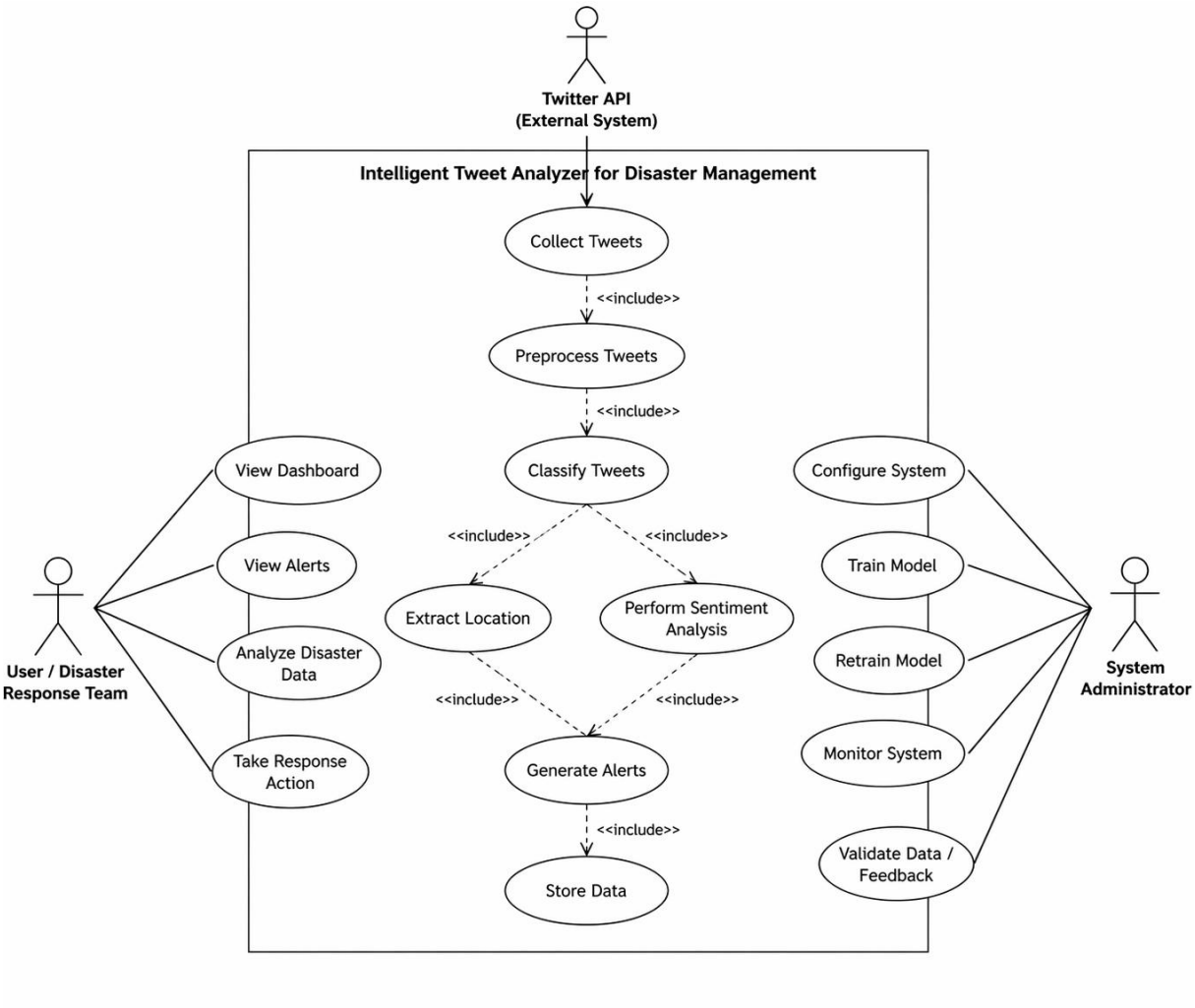

Fig 5.3 Use Case Diagram

Actors:

- System Administrator / Data Engineer
- Disaster Response Personnel / Analyst / User
- External Data Source (Twitter / Social API)
- Use Cases:
- Collect Tweets
- Clean / Filter Tweets
- Train Classification Model
- Classify Tweets
- Extract Location
- Generate Alert

- View Dashboard / Map
- Provide Feedback (label correction)
- Retrain Model

5.4 Sequence Diagram

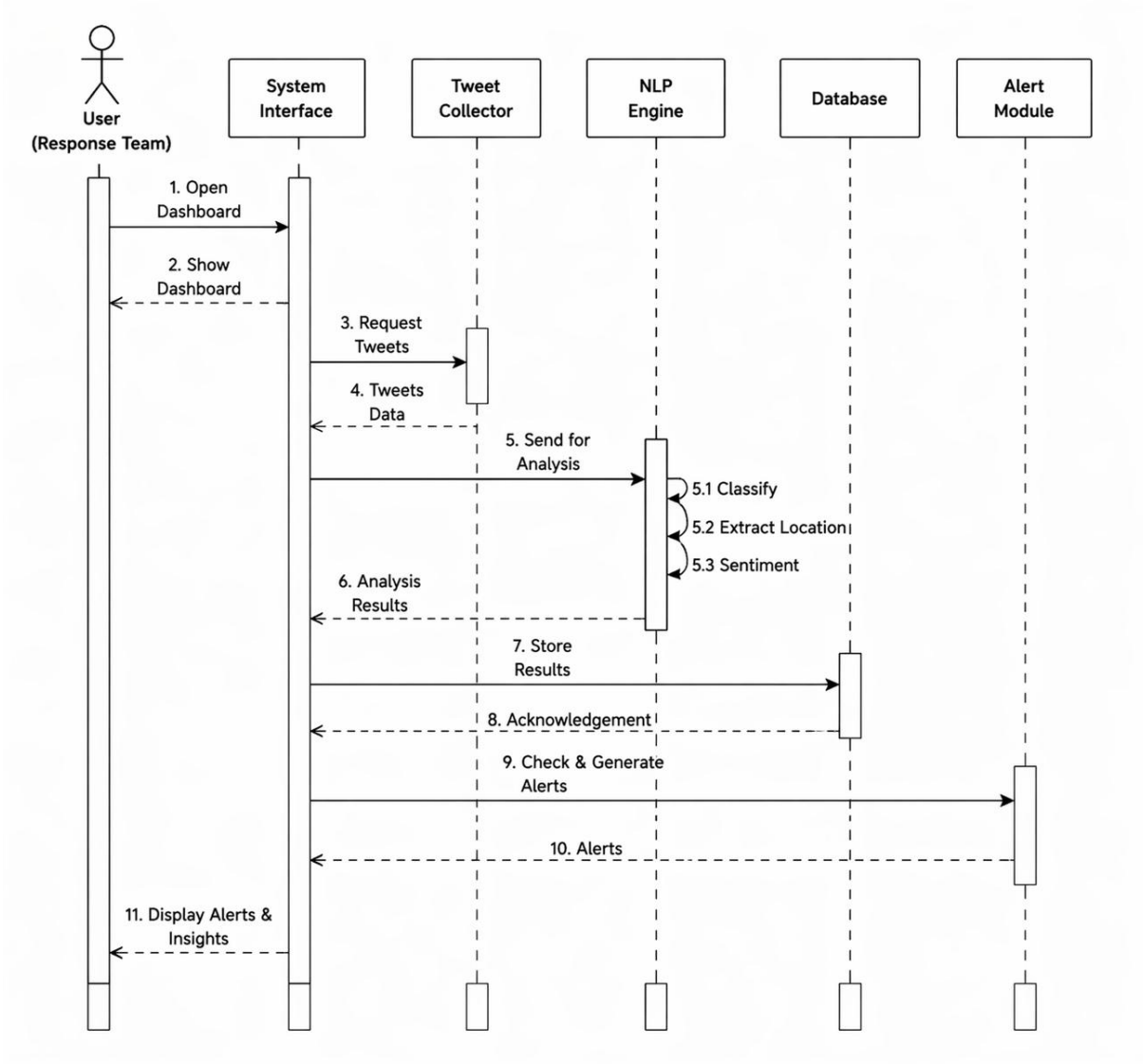

Fig 5.4 Sequence Diagram

Sequence:

- Twitter API → sends tweet to Tweet Collector
- Tweet Collector → sends the tweet to Preprocessor
- Preprocessor → returns cleaned/filtered tweet to Classifier
- Classifier → returns classification (e.g. “flood, high severity”)
- Classifier → sends to Geolocation Module (maybe concurrently)
- Geolocation Module → returns location coordinates

- Classifier + Location → sent to Alert Engine
- Alert Engine → decides whether to trigger alert
- If yes, Alert Engine → sends alert info to Dashboard / UI
- Dashboard updates the map, shows alert to user

Analysis:

Strengths:

- Modular architecture (clear separation of frontend, backend, AI, and data).
- Real-time disaster monitoring via RSS and AI analysis.
- Scalable REST API with modern Python stack.
- Extensible for new data sources or analytics.

Weaknesses:

- Reliance on external RSS feeds (may be blocked or rate-limited).
- AI analysis latency (Gemini API response time).
- CORS/configuration issues may affect frontend-backend integration.

Opportunities:

- Add more data sources (social media, sensors).
- Enhance analytics (trend prediction, geospatial clustering).
- Improve alerting (push notifications, SMS integration).

Threats:

- API changes or deprecation (Gemini, RSS).
- Data privacy and security concerns.
- Scalability under high load.

5.5 Activity Diagram

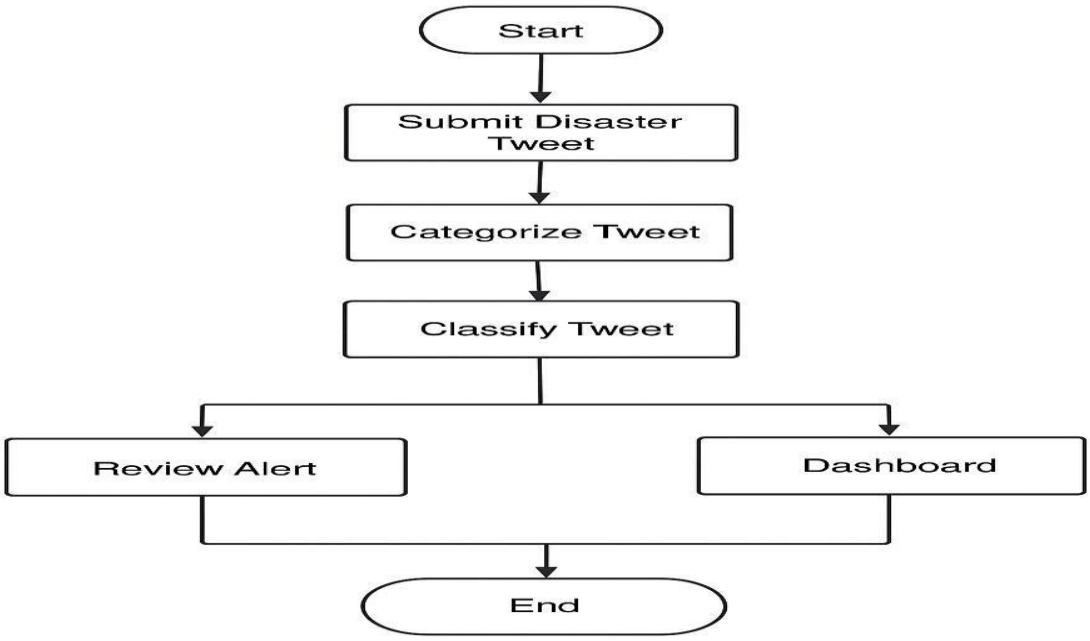

Fig 5.5 Activity Diagram

The Activity Diagram represents the dynamic workflow of the Tweet Analysis for Disaster Management System, showing the sequence of operations performed from data collection to disaster response decision-making. It highlights how raw tweet data moves through the system, the processing stages involved, and the actions taken by users and the system at each step.

Step-by-Step Workflow Description

Start –
The process begins when the system is activated to monitor disaster-related tweets in real time.

Tweet Collection -

The system connects to the Twitter API and fetches tweets using predefined disaster- related keywords (e.g., flood, earthquake, rescue) and location filters.

DataStorage –

The collected tweets are temporarily stored in the Raw Tweet Database for preprocessing and further analysis.

Data Preprocessing –

The system performs text cleaning and normalization, including removal of URLs, stopwords, hashtags, emojis, and special characters.

Decision Node: If the tweet is invalid or empty after cleaning → it is discarded.

Otherwise → it proceeds to the next step.

Feature Extraction –

The cleaned tweets are tokenized and converted into numerical vectors using techniques like TF- IDF or word embeddings for machine learning analysis.

1. **Tweet Classification (Core Analysis)** –

The preprocessed data is passed to the Google Gemini or an ML model (BiLSTM-CNN or Random Forest) to classify each tweet into one of the following categories:

- Help Request
- Rescue Update
- Blood Donation
- Resource Availability
- General Information

2. **Entity Extraction (NER)** –

   The model identifies key entities (locations, victim counts, resources like "food" or "ambulance") to add context to each tweet.

3. **Store Classified Data** –

   The results are stored in the Classified Data Repository, tagged with labels and urgency scores.

4. **Analytics and Visualization** –

   The Analytics Engine aggregates data to generate insights such as sentiment trends, disaster hotspots, and resource requirements.

These insights are displayed on an interactive Dashboard featuring maps, charts, and real-time analytics.

5. **Alert Generation –**

   If a tweet is classified as urgent (e.g., life-threatening situations), the system automatically generates alerts for first responders or NGOs.

6. **Response Action –**

   The Disaster Management Team or Rescue Workers receive the alert and take appropriate action (dispatching rescue teams, allocating resources, etc.).

7. **End –**

   The activity concludes once data is stored, visualized, and necessary actions have been initiated.

Key Components of the Proposed Framework:

- Frontend (React): User interface, data visualization, map, analytics, alert display.
- Backend (FastAPI): REST API, incident processing, analytics, alert generation, CORS, RSS monitoring.
- Database (MongoDB): Stores incidents, alerts, analytics summaries.
- Gemini Agent: AI-powered content analysis for incident relevance, severity, and alert generation.
- RSS Monitor: Periodically fetches and processes disaster-related RSS feeds.
- Models (Pydantic): Data validation and serialization for API responses.

Analysis:

Strengths:

- Modular architecture (clear separation of frontend, backend, AI, and data).
- Real-time disaster monitoring via RSS and AI analysis.
- Scalable REST API with modern Python stack.
- Extensible for new data sources or analytics.

Weaknesses:

- Reliance on external RSS feeds (may be blocked or rate-limited).
- AI analysis latency (Gemini API response time).

- CORS/configuration issues may affect frontend-backend integration.

Opportunities:

- Add more data sources (social media, sensors).
- Enhance analytics (trend prediction, geospatial clustering).
- Improve alerting (push notifications, SMS integration).

Threats:

- API changes or deprecation (Gemini, RSS).
- Data privacy and security concerns.
- Scalability under high load.

Chapter 6

Experimental Setup

6. Experimental Setup

6.1 Software Requirements

- Operating System: Windows 10/11, Linux (Ubuntu 22.04 LTS recommended)
- Programming Languages: Python 3.10+, JavaScript (for front-end)
- Web Frameworks: FastAPI (backend API), React.js (frontend)
- Database: MongoDB (for raw and processed tweet data)
- Visualization Tools: Plotly, Matplotlib, Seaborn (for charts and heatmaps)
- APIs: Twitter API (data collection), Google Gemini API (AI/NLP processing), Google Maps API (geolocation)
- Development Tools: Visual Studio Code, GitHub Codespaces, Jupyter Notebook
- Version Control: Git & GitHub for collaborative development and version tracking

6.2 Hardware Requirements

Minimum Configuration:

- Processor: Dual-Core CPU (2.0 GHz or above) RAM: 8 GB

- Storage: 10 GB free space

Recommended Configuration:

- Processor: Quad-Core CPU (Intel i5/i7 or AMD equivalent) RAM: 16 GB or higher
- Storage: 100 GB SSD
- Internet: Stable broadband connection for real-time tweet streaming and API calls

Devices Used: Laptop, Desktop, and Smartphone (for dashboard monitoring and mobile alerts)

6.3 Development Environment

- Platform: GitHub Codespaces (hosted on Microsoft Azure Virtual Machine)
- Environment Details: Linux OS with 4-core CPU, 16 GB RAM, pre-configured Python and Node.js
- Containerization: Docker used for environment consistency across development and testing stages
- Agents & AI Assistants: GitHub Copilot and Emergent AI for assisted coding, debugging, and documentation
- Versioning: Each system module (frontend, backend, ML model) maintained in independent Git branches for modular deployment
- CI/CD Pipeline: GitHub Actions for automated testing, integration, and deployment to staging environment

6.4 ML/NLP Libraries and Frameworks

- Google Gemini LLM: Core NLP model for text classification, sentiment analysis, and named entity recognition (NER)
- Google Generative AI: Used for text summarization and semantic understanding
- Scikit-learn: Classical machine learning algorithms (Random Forest, SVM, Logistic Regression)
- TensorFlow / PyTorch: For deep learning-based classification and fine-tuning
- Hugging Face Transformers: For tokenization, embedding generation, and pre-trained NLP models
- NLTK / SpaCy: For text cleaning, lemmatization, and stopword removal
- Pandas / NumPy: For structured data manipulation and analysis

- Matplotlib / Seaborn: For generating performance visualizations and analytics dashboards

6.5 Implementation

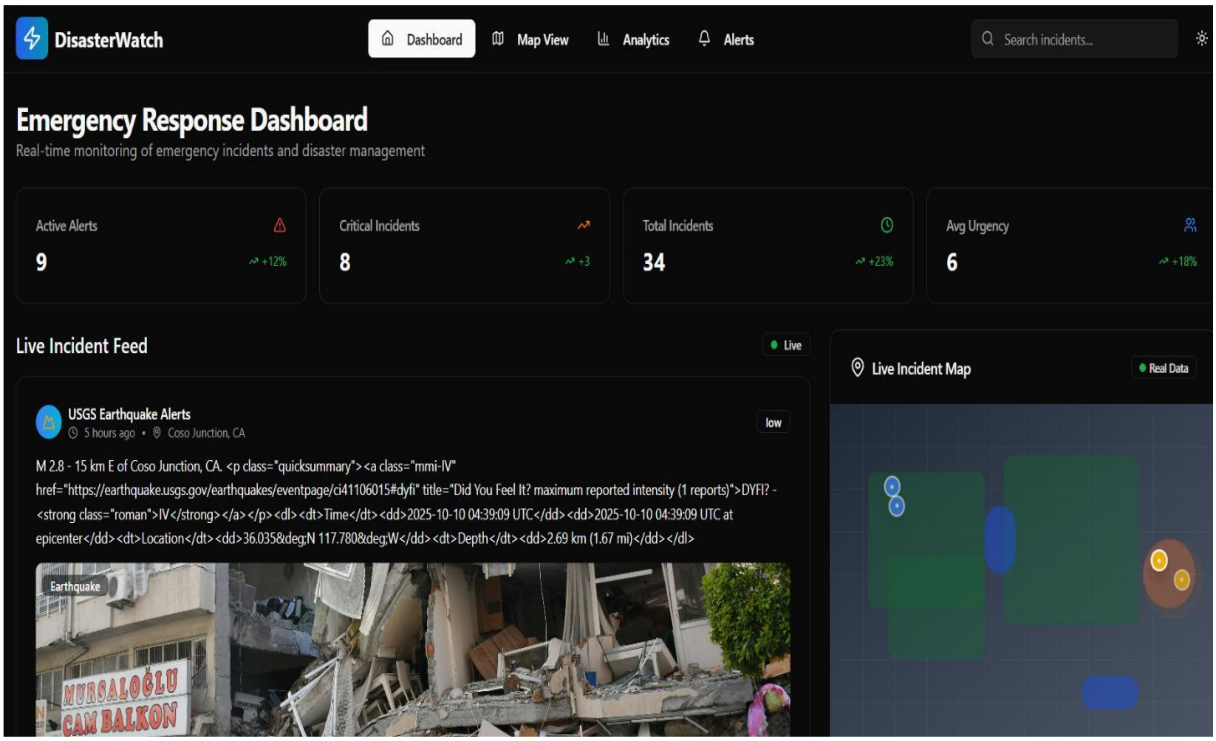

Fig 6.5.1 Dashboard

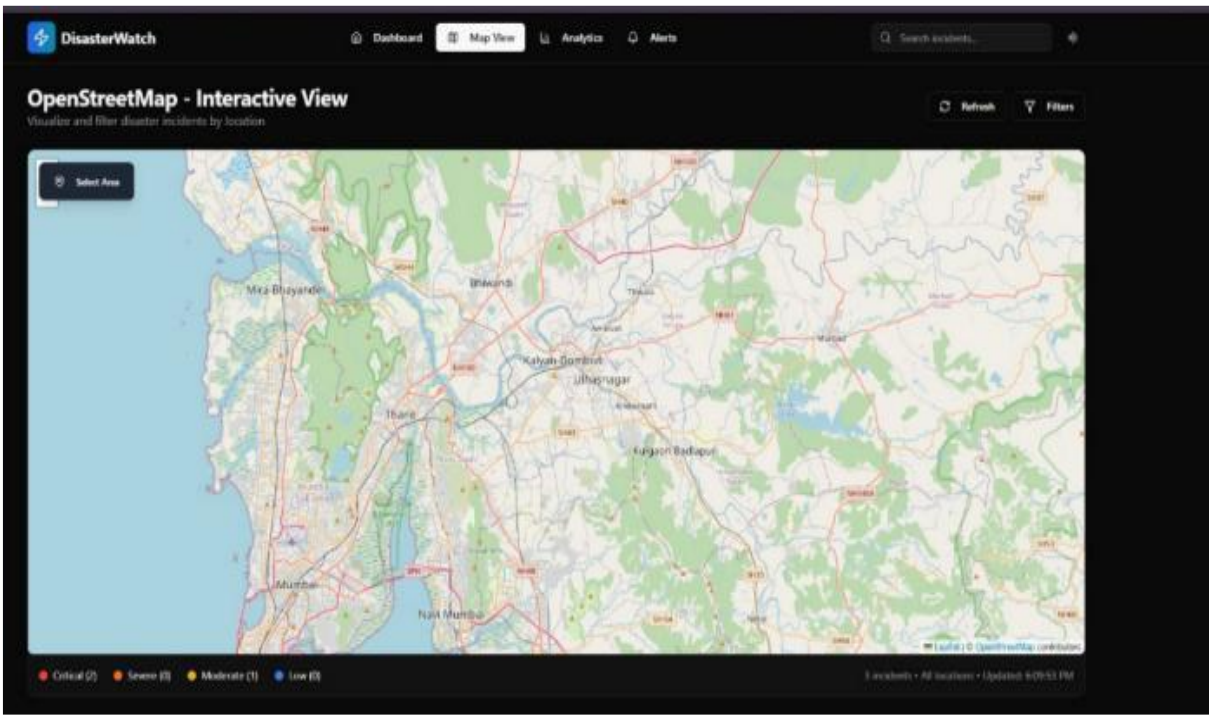

Fig 6.5.2 Map View

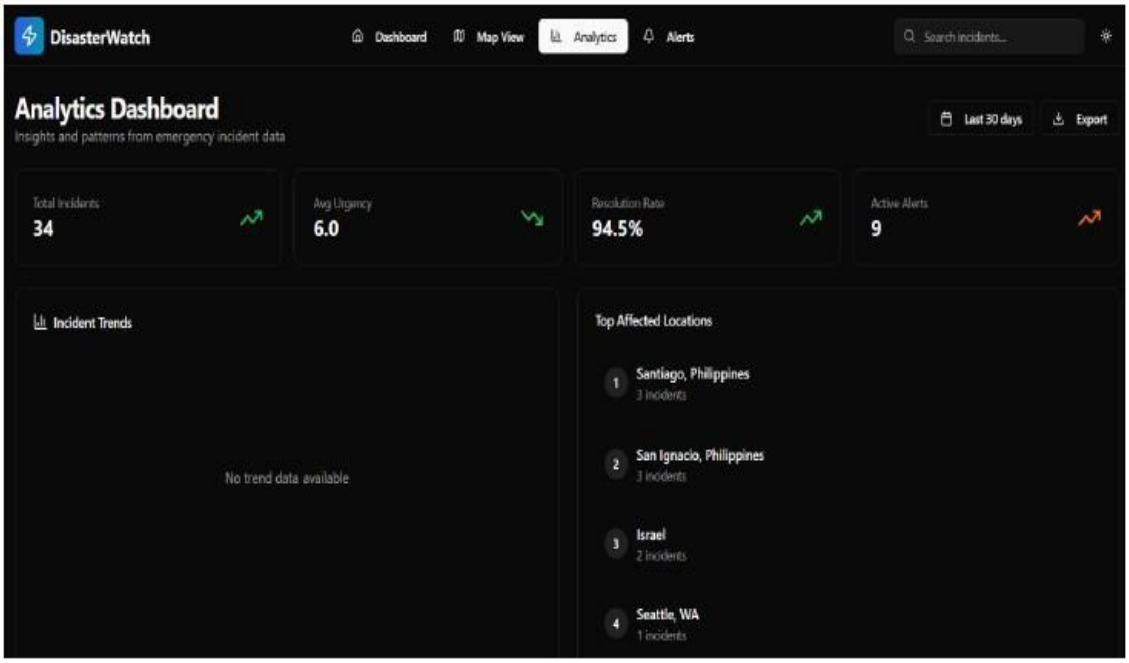

Fig 6.5.3 Analytics

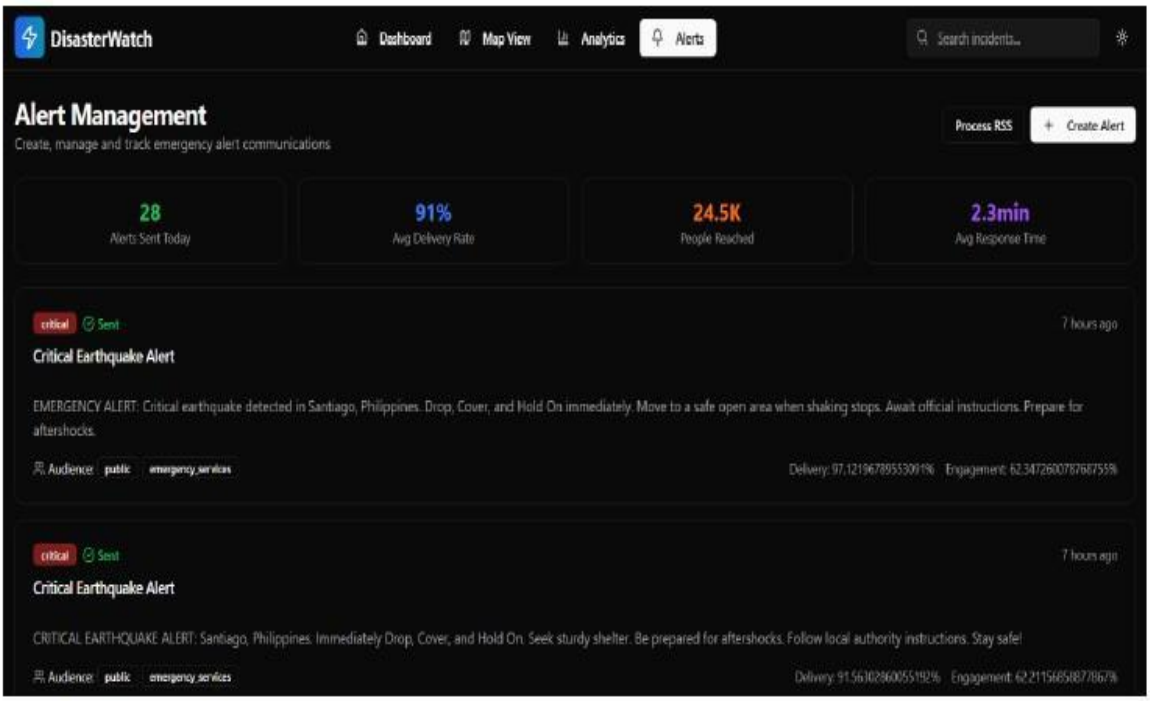

Fig 6.5.4 Alerts

Chapter 7

Result and Discussion

7. Result and Discussion

7.1 Result

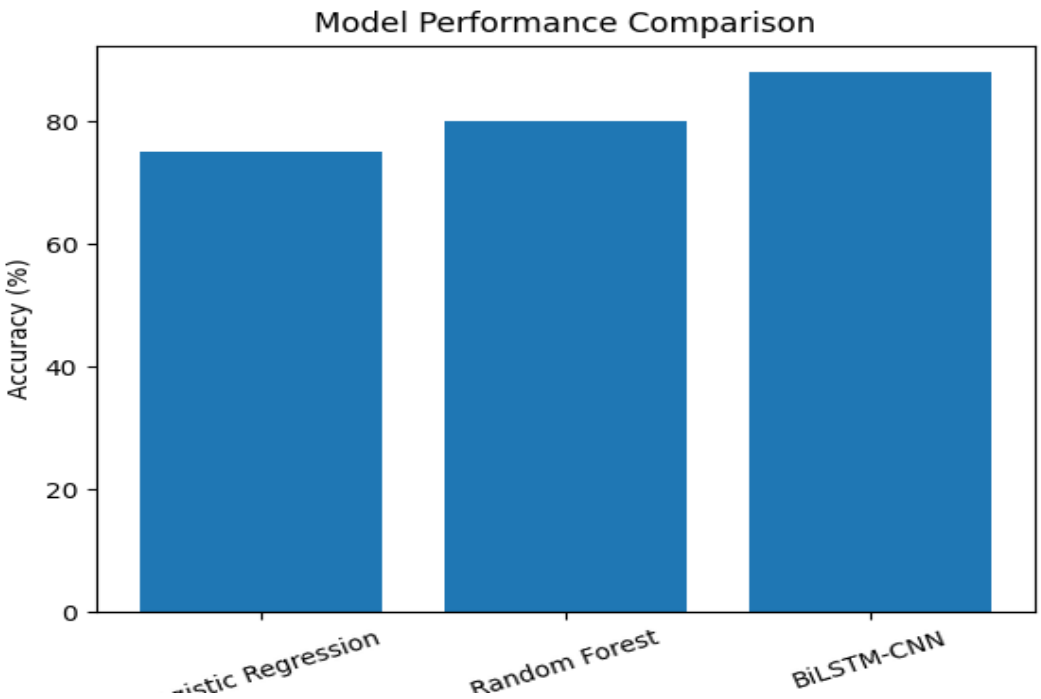

Fig 7.1 model performance comparison

This graph illustrates the comparison of different machine learning models used for tweet classification, including Logistic Regression, Random Forest, and the proposed deep learning model (BiLSTM-CNN). It is observed that the deep learning model achieves the highest accuracy, around **88%**, due to its ability to capture contextual and sequential patterns in textual data. Traditional models show comparatively lower performance as they rely on simpler feature representations. This demonstrates the effectiveness of deep learning approaches in handling complex and unstructured social media data.

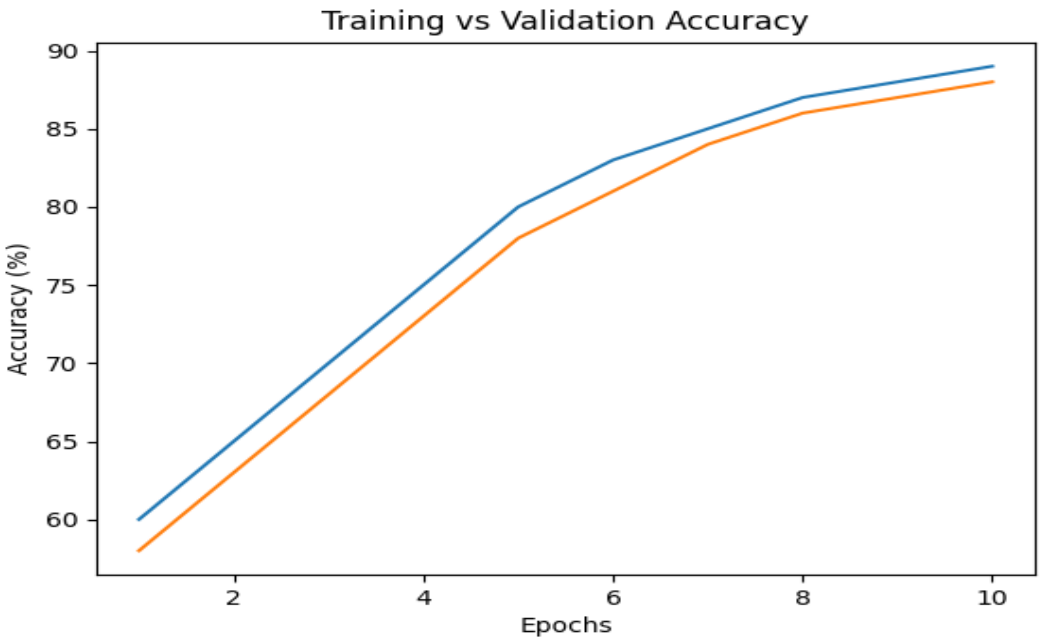

Fig 7.2 Training vs Validation Accuracy

This graph represents the training and validation accuracy of the model over multiple epochs. Both curves show a steady increase, indicating that the model is learning effectively from the dataset. The small gap between training and validation accuracy suggests that the model does not suffer from significant overfitting and generalizes well to unseen data. This stability confirms that the chosen model architecture and training strategy are appropriate for the problem.

Fig 7.3 Class-wise precision

The graph shows precision values for each tweet category such as Help Requests, Rescue Updates, Blood Donations, Resource Availability, and General Information. Higher precision in categories like Help Requests and Blood Donations indicates that the model is highly accurate in predicting critical tweets without producing many false positives. This is particularly important in disaster scenarios where incorrect classification can lead to misallocation of resources.

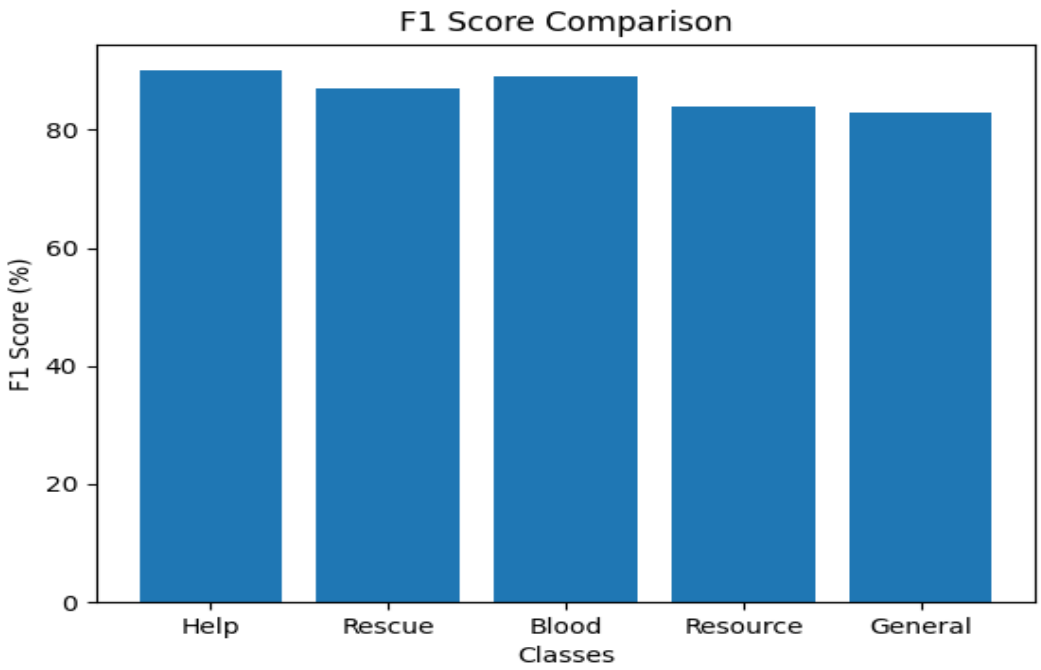

Fig 7.4 F1 Score

The F1-score graph combines both precision and recall to provide a balanced evaluation of model performance. The results show that the model maintains consistently high F1-scores across all categories, indicating reliable classification performance. The balanced scores demonstrate that the model is neither overfitting nor biased toward any particular class, making it robust for real-world deployment

7.2 Discussion

The results show that the proposed Tweet Analysis system works effectively in identifying and classifying disaster-related tweets. The model achieved good accuracy and was able to correctly detect important categories such as help requests and rescue updates. This helps in quickly identifying critical situations during disasters.

The system also performs well in handling large amounts of unstructured data. By using preprocessing techniques, it removes noise like emojis, links, and irrelevant text, which improves the overall performance of the model. The use of machine learning and NLP helps the system understand the meaning of tweets more accurately.

Another important feature is urgency detection and sentiment analysis, which helps in prioritizing tweets based on their importance. This ensures that emergency cases are handled first. The geolocation feature also helps in identifying affected areas, which can support better decision-making during disaster response.

However, the system has some limitations. It may not always perform well with slang, sarcasm, or multilingual tweets. Also, real-time processing requires good computational resources. Despite these challenges, the system is efficient and useful for disaster management.

Overall, the proposed system improves speed, accuracy, and automation compared to existing methods, making it a helpful tool for real-time disaster response.

Chapter 8

Conclusion and Future Work

8. Conclusion and Future Work

The Intelligent Tweet Analyzer for Disaster Management project demonstrates the effective use of social media data, particularly Twitter (X), for improving disaster response and management. The system successfully leverages Natural Language Processing (NLP) and Machine Learning techniques to analyze large volumes of real-time tweets and extract meaningful, actionable insights. By classifying tweets into categories such as Help Requests, Rescue Updates, Blood Donations, and General Information, the system enables faster identification of critical situations.

The integration of sentiment analysis, urgency detection, and geolocation tagging further enhances the system's ability to prioritize high-risk cases and support decision-making for disaster response teams. The proposed architecture, consisting of data collection, preprocessing, classification, analytics, and alert generation modules, ensures a structured and scalable approach to handling unstructured social media data.

Overall, the system helps bridge the gap between real-time social media information and practical disaster management operations. It improves situational awareness, reduces response time, and supports efficient resource allocation. Despite challenges such as noisy data, multilingual content, and computational requirements, the system proves to be a valuable tool for enhancing emergency response and saving lives.

Although the proposed system provides an effective solution, there are several areas for future enhancement:

1.Multilingual Support: Extend the system to handle multiple regional languages (e.g., Hindi, Marathi, Tamil) to improve inclusivity and coverage in diverse regions.

2. Multimodal Analysis: Incorporate image and video analysis along with text to improve disaster detection and situational understanding.

3.Advanced Deep Learning Models: Integrate more advanced models like transformers (BERT, GPT-based models) for higher accuracy and contextual understanding.

4.Real-Time Scalability: Optimize the system using distributed computing and cloud technologies to handle large-scale real-time tweet streams efficiently.

5.Integration with Government Systems: Connect the system with official disaster management platforms and emergency services for direct action and coordination.

6.Mobile Application Development: Develop a mobile app for real-time alerts and easier access for responders and the general public.

7.Fake News Detection: Add modules to detect misinformation or fake tweets to ensure reliability of extracted information.

8.Predictive Analytics: Use historical data and AI models to predict disaster trends and potential high-risk areas in advance.

9.Improved Geolocation Accuracy: Enhance location extraction using advanced GIS techniques and external APIs for precise mapping.

10.User Feedback and Model Retraining: Implement feedback mechanisms to continuously improve model accuracy through retraining.

References

[1] Arvandi, Alireza, J. Rokne and A. Reda, Twitter analysis in emergency management: recent research and trends, 2024.

[2] Karimiziarani, Mohammadsepehr and . H. Moradkhani, Social response and Disaster management: Insights from twitter data Assimilation on Hurricane Ian, 2023.

[3] K. M. Yasin and S. Madria, A deep learning approach for tweet classification and rescue scheduling for effective disaster management, 2019.

[4] Nair, M. R, G. R. , Ramya and P. B. Sivakuma, Usage and analysis of Twitter during 2015 Chennai flood towards disaster management, 2017.

[5] S. H. and S. Setty, Disaster Analysis Through Tweets, 2015.

[6] M. . G. and R. K. Dhanaraj, A multi-model approach for disaster-related tweets classification, 2024.

[7] L. and . A. Duc, Disaster tweets classification using BERT-based language model, 2022.

[8] Z. H. M., R. Jalota, M. A. Sherif, and A.-C. N. Ngomo, I-AID: identifying actionable information from disaster-related tweets, 2020.

[9] Z. H. Peng, Y. Zhou, C. Caragea and . D. Caragea, Crisismatch: Semi-supervised few-shot learning for fine-grained disaster tweet classification, 2023.

[10] B. M. B. Alam, Z. Fatima and S. Shaikh, Natural disaster tweets classification using multimodal data, 2023.

[11] V. S. A. L, H. K. Starbird and L. Palen, Microblogging during two natural hazards events: what twitter may contribute to situational awareness, 2010.

[12] Starbird, K. and L. Palen, "Voluntweeters": Self-Organizing by Digital Volunteers in Times of Crisis, 2011.

[13] I. M. C. Castillo, F. Diaz and S. Vieweg, Processing social media messages in mass emergency: A survey, 2015.

[14] O. A. . C. Castillo, F. Diaz and S. Vieweg, Crisislex: A lexicon for collecting and filtering microblogged communications in crises, 2014.

[15] P. . H. A. Hampton, S. Bhatt, V. L. Shalin, A. P. Sheth and J. M. Flach, Identifying seekers and suppliers in social media communities to support crisis coordination, 2014.

[16] R. . C. A. Marx and V. Pipek, Crisis management 2.0: Towards a systematization of social software use in crisis situations, 2012.

[17] S. T. M. Okazaki and Y. Matsuo, Earthquake shakes twitter users: real-time event detection by social sensors, 2010.

[18] C. A. A. Croitoru, . A. Stefanidis and J. Radzikowski, # Earthquake: Twitter as a distributed sensor system, 2013.

[19] E. P. S, D. C. Bowden and M. Guy, Twitter earthquake detection: earthquake monitoring in a social world, 2011.

[20] M. S. E, L. Middleton and S. Modafferi, Real-time crisis mapping of natural disasters using social media, 2013.

Appendix

A. Setup Instructions

A.1 System Requirements

Ensure the following software is installed:

- Python: Version 3.10 or above (Recommended: 3.11)
- Node.js: Version ≥ 18 (Recommended: 20 LTS)
- npm / yarn
- Git
- MongoDB

A.2 Cloning the Repository

Step 1: Clone the repository

$ git clone https://github.com/your-repo/tweet-analyzer-disaster-management

Step 2: Navigate to project folder

$ cd tweet-analyzer-disaster-management

A.3 Frontend Setup (React)

Step 1: Navigate to client folder

$ *cd client*

Step 2: Install dependencies

$ *npm install*

Step 3: Run frontend

$ *npm start*

Frontend runs at:
http://localhost:3000

A.4 Backend Setup (FastAPI Server)

Step 1: Navigate to server folder

$ *cd backend*

Step 2: Install required python libraries

$ pip install -r requirements.txt

Step 3: Run server

$ uvicorn main:app --port 8000 –reload

Backend runs at:

http://127.0.0.1:8000

A.5 Databse Setup (MongoDB)

Step 1: Navigate to backend folder

$ *mongod*

Step 2: Connect databse in project

ConfigureMongoDB URL in .env file

MONGO_URI=mongodb://localhost:27017/disaster_db

A.6 Running the Complete System

Step 1: Create developer account

- Register at Twitter Developer Portal

Step 2: Generate API keys

- API Key

- API Secret
- Bearer Token

Step 3: Add keys to environment file

A.7 Important Notes

To run the full system:

1.Start Frontend (React)

2. Start FastAPI Backend

3.Start MongoDB

All services must run simultaneously for full functionality**.**

A.8 Important Notes

Ensure MongoDB service is running

Ensure ports (3000, 8000) are free

Internet is require for: Twitter API, Maps

A.9 Setup(Virtual Environment)

Create virtual environment:

python -m venv venv

Activate environment:

venv\Scripts\activate

Publication

Submission Summary

Conference Name
3rd International Conference on Business Intelligence and Data Analytics

Paper ID
970

Paper Title
Intelligent Tweet Analyzer for Disaster Management

Abstract
In recent years, social media platforms such as Twitter (now X) have emerged as critical tool for real-time information
exchange during disaster situations. This project, Tweet Analysis for Disaster Management, aims to harness the potential
of Twitter data to support faster, data driven decision-making during crises like floods, earthquakes, and pandemics. The
system utilizes Natural Language Processing (NLP) and Machine Learning (ML) techniques to automatically analyze
tweets, extract relevant information, and categorize them into actionable types such as Help Requests, Rescue Updates,
Blood Donations, and General Information. By integrating geolocation tagging, sentiment analysis, and urgency scoring,
the proposed framework enhances situational awareness and enables disaster response teams to prioritize resources
effectively. The architecture combines real-time data collection, AI-powered tweet classification, and visualization modules
to support timely and targeted interventions. The system addresses challenges such as noisy and multilingual tweet data,
ensuring scalability and adaptability across diverse disaster scenarios. Overall, this approach bridges the gap between
social media intelligence and operational disaster management, improving response efficiency, coordination, and lifesaving outcomes.

Created
10/02/2026, 00:02:42

Last Modified
10/02/2026, 00:02:42

Authors
Hetal Mahajan (AP Shah Institute Of Technology) <hetalmahajan355@apsit.edu.in>
Prabudh Gaikwad (AP Shah Institute Of Technology) <prabhudhgaikwad377@apsit.edu.in>
Emails sent.
P Shah Institute Of Technology) <chrsitinadcruz23@apsit.edu.in>
Dr. Rahul Ambekar (AP Shah Institute Of Technology) <rkambekar@apsit.edu.in>

Intelligent Tweet Analyzer for Disaster Management

Hetal Mahajan
Department of Computer Science and Engineering (AI-ML)
A.P Shah Institute of Technology
Thane, Maharashtra
hetalmahajan355@apsit.edu.in

Prabudh Gaikwad
Department of Computer Science and Engineering (AI-ML)
A.P Shah Institute of Technology
Thane, Maharashtra
prabhudhgaikwad377@apsit.edu.in

Christina Dcruz
Department of Computer Science and Engineering (AI-ML)
A.P Shah Institute of Technology
Thane, Maharashtra
christinadcruz23@apsit.edu.in

Dr. Rahul K Ambekar
Department of Computer Science and Engineering (AI-ML)
A.P Shah Institute of Technology
Thane, Maharashtra
rkambekar@apsit.edu.in

Abstract:

In recent years, social media platforms such as Twitter (now X) have emerged as critical tool for real-time information exchange during disaster situations. This project, Tweet Analysis for Disaster Management, aims to harness the potential of Twitter data to support faster, data driven decision-making during crises like floods, earthquakes, and pandemics. The system utilizes Natural Language Processing (NLP) and Machine Learning (ML) techniques to automatically analyze tweets, extract relevant information, and categorize them into actionable types such as Help Requests, Rescue Updates, Blood Donations, and General Information. By integrating geolocation tagging, sentiment analysis, and urgency scoring, the proposed framework enhances situational awareness and enables disaster response teams to prioritize resources effectively. The architecture combines real-time data collection, AI-powered tweet classification, and visualization modules to support timely and targeted interventions. The system addresses challenges such as noisy and multilingual tweet data, ensuring scalability and adaptability across diverse disaster scenarios. Overall, this approach bridges the gap between social media intelligence and operational disaster management, improving response efficiency, coordination, and life-saving outcomes.

Keywords: Disaster Management, Twitter Analysis, NLP, Machine Learning, Sentiment Analysis, Real time responses, Geolocation, Emergency Classification

1. INTRODUCTION

Social media platforms, particularly Twitter (now X), have become vital real-time information channels during disaster situations, offering a dynamic and immediate source of data that traditional methods, such as news reports or official emergency channels, often cannot match in speed or granularity. In crises like hurricanes, earthquakes, floods, or pandemics, individuals use Twitter to share critical updates, report on-the-ground conditions, request urgent assistance (e.g., medical aid, food, or shelter), offer support (e.g., donations or volunteer services), and disseminate situational awareness (e.g., road closures or safe zones). This user-generated content provides a wealth of real-time, geospatially relevant information that can significantly enhance disaster management efforts, enabling faster decision-making and resource allocation to save lives and mitigate impacts.

However, the sheer volume, velocity, and variety of tweets during disasters present significant challenges. Millions of tweets may be posted within hours, as seen during Hurricane Ian (2023), where over 20 million tweets were analyzed to understand public responses. These tweets are often noisy, containing informal language, slang, emojis, or irrelevant content, which complicates the identification of urgent needs. Manual analysis of such data is infeasible due to the scale and time sensitivity, leading to delays in recognizing critical information, such as requests for rescue or medical supplies. Additionally, the diversity of tweet content ranging from emotional outbursts to actionable updates further hinders the ability of first responders and non-governmental organizations (NGOs) to prioritize and act on the most pressing needs.

To address these challenges, the proposed solution is to develop an automated system that leverages advanced natural language processing (NLP) and machine learning techniques to classify tweets into actionable categories, such as Help Request, Blood Donation, Rescue Update, Resource Availability, and General Information. This system would employ algorithms like sentiment analysis, topic modelling, and deep learning models (e.g., BiLSTM,CNN, as used in the 2019 study on Hurricanes Harvey and Irma) to filter and categorize tweets with high accuracy. For example, a tweet stating, "Urgent: Need blood type O+ in Chennai hospital" could be classified as a Blood Donation request, while "Rescue team reached Main Street" could be tagged as a Rescue Update. By incorporating geolocation data and sentiment analysis, as demonstrated in the Chennai flood study (2017), the system can prioritize tweets based on urgency and location, enabling targeted responses.

The impact of such a system is profound for disaster management. By automating the classification of tweets, it enables first responders, NGOs, and government agencies to quickly identify and prioritize urgent needs, such as dispatching rescue teams to stranded individuals or directing medical supplies to areas with critical shortages. This enhances coordination, reduces response times, and optimizes resource allocation, ultimately saving lives and improving efficiency. For instance, Shekhar and Setty (2015) showed how sentiment-based visualization of tweets can map distress levels, aiding responders in focusing on high-priority areas. Furthermore, by integrating with existing disaster management frameworks, the system can bridge the gap between social media insights and operational workflows, addressing a key limitation noted by Arvandi et al. (2024). Despite challenges like noisy data, multilingual content, and computational demands, as highlighted in the literature, this solution offers a scalable, real-time approach to harness Twitter's potential, making it a game-changer for effective disaster response.

Real-Time Information Hub: Social media platforms, particularly Twitter (now X), serve as critical channels during disasters like hurricanes, earthquakes, floods, or pandemics, providing immediate, user-generated updates that outpace traditional sources like news or official reports.

User Contributions: Individuals post about emergencies (e.g., stranded locations), request help (e.g., medical aid, food), offer support (e.g., donations, volunteering), and share updates (e.g., road closures, safe zones), offering valuable on-the ground insights.

Sustainability Development Goal Alignment

- SDG 11: Sustainable Cities and Communities

The system enhances the ability of cities and communities to prepare for, respond to, and recover from disasters.

By analyzing tweets for real-time information on emergencies, it helps authorities coordinate rescue operations and resource allocation efficiently, contributing to disaster-resilient infrastructure and sustainable urban management.

- SDG 9: Industry, Innovation, and Infrastructure

The system demonstrates innovative use of emerging technologies like Natural Language Processing (NLP), Machine Learning (ML), and Large Language Models (LLMs) for societal benefit. It promotes the integration of digital infrastructure in emergency management systems.

II. LITERATURE SURVEY

Social media–driven disaster management has gained significant attention in recent years due to the rapid growth of platforms such as Twitter (now X) and their ability to provide real-time, user-generated information during emergencies. A. Arvandi et al. [1] presented a comprehensive survey on Twitter analysis in emergency management, highlighting the evolution of sentiment analysis, topic modelling, and geolocation-based techniques for understanding public response during disasters. Their study emphasized the importance of social media as a complementary data source to traditional disaster management systems. However, the authors identified critical gaps, including limited multilingual tweet analysis and insufficient integration with operational disaster response frameworks, which restrict real-world applicability.

Karimiziarani and Moradkhani [2] analyzed public sentiment and social response during Hurricane Ian using over 20 million tweets. Their work employed natural language processing and spatiotemporal analysis to examine disaster evolution and population behavior. While the large-scale dataset provided valuable insights into disaster dynamics, the study suffered from challenges related to noisy and irrelevant

tweets, which reduced the accuracy of spatial interpretations. Additionally, the absence of multimodal data analysis, such as images and videos, limited the system's ability to capture the full context of disaster situations.

Deep learning approaches have also been explored for improving tweet classification accuracy in disaster scenarios. Kabir and Madria [3] proposed an attention-based Bi LSTM and CNN hybrid model for tweet classification and rescue scheduling during Hurricanes Harvey and Irma. Their system demonstrated improved performance in identifying actionable tweets such as rescue requests. Despite these advancements, the approach relied heavily on labeled datasets and required substantial computational resources, making it less suitable for real-time deployment in resource-constrained environments.

Earlier research by Nair et al. [4] examined Twitter usage during the 2015 Chennai floods to understand information propagation and public communication patterns. Using traditional machine learning algorithms such as Random Forests, the study confirmed the effectiveness of Twitter for real-time disaster communication. However, it largely ignored non-English tweets, particularly Tamil-language content, and lacked integration with real-world disaster response systems, reducing its practical utility in linguistically diverse regions.

Shekhar and Setty [5] explored sentiment classification and visualization techniques to analyze disaster-related tweets across multiple events. Their work demonstrated how sentiment trends could be mapped geographically to identify high-distress areas. While the study laid foundational work for disaster sentiment analysis, it focused primarily on emotional polarity rather than actionable intelligence and did not generalize well across different disaster types.

Imran *et al.* [6] conducted a detailed survey on processing social media messages during mass emergency events, reviewing techniques for information extraction, classification, and credibility assessment. The study highlighted key challenges such as data noise, scalability, and misinformation. Although the survey provided a strong theoretical foundation, it emphasized the need for real-time, action-oriented systems that directly support emergency responders, a gap that motivates the proposed work.

III. LIMITATIONS OF EXISTING SYSTEM

Although social media–based disaster management systems have demonstrated considerable potential, existing approaches suffer from several critical limitations that hinder their effectiveness in real-world emergency response scenarios. One of the primary challenges is the noisy and unstructured nature of Twitter data. Tweets frequently contain informal language, abbreviations, spelling errors, emojis, hashtags, URLs, and sarcasm, which complicate accurate natural language understanding. Many existing systems struggle to differentiate between genuine emergency requests and non-actionable content such as opinions, rumors, or emotional expressions, leading to misclassification and reduced reliability during critical situations.

Most models are trained on datasets from specific disaster events such as hurricanes or floods and perform poorly when applied to different disaster scenarios like earthquakes, wildfires, or pandemics. This dependency on disaster-specific labeled data necessitates frequent retraining and fine-tuning, which is impractical during rapidly evolving emergencies. Consequently, system scalability and adaptability remain major concerns. Language bias is another persistent challenge. A large proportion of existing systems are optimized primarily for English-language tweets, neglecting multilingual and code-mixed content commonly observed in regions with linguistic diversity. This exclusion of non-English tweets leads to incomplete situational awareness and potentially overlooks critical distress signals from marginalized or rural populations, thereby reducing equity and effectiveness in disaster response.

Real-time processing and scalability also pose serious limitations. During large-scale disasters, millions of tweets may be generated within a short time frame. Many existing systems lack the computational efficiency and infrastructure required to process such high-velocity data streams in real time. High latency in tweet analysis can delay the identification of urgent situations, directly impacting response time and resource allocation. Finally, most existing systems exhibit poor integration with operational disaster management frameworks. Outputs are often limited to offline analysis, research dashboards, or post-event assessments, with minimal linkage to alerting systems, emergency services, or decision-support tools used by first responders and NGOs. This disconnect between analytics and action significantly reduces the real-world impact of social media–based disaster analysis systems.

IV. PROBLEM STATEMENT AND OBJECTIVES

A. PROBLEM STATEMENT:

During disaster situations, Twitter generates vast amounts of real-time information that can significantly aid emergency response efforts. However, due to the unstructured and noisy nature of tweets, authorities struggle to identify urgent and actionable information promptly. Existing systems fail to effectively combine classification, sentiment analysis, urgency assessment, and location extraction in a unified real-time framework. This results in delayed response, inefficient resource allocation, and missed opportunities to save lives.

B. OBJECTIVES:

- To design an automated pipeline for real-time collection and preprocessing of disaster-related tweets.
- To classify tweets into actionable emergency categories using ML and NLP techniques.
- To perform sentiment analysis and urgency scoring for prioritization of critical cases.
- To extract and tag geographic entities to support spatial analysis and targeted response.
- To provide decision-support dashboards and alerts for disaster response agencies.

V. METHODOLOGY

The proposed system for real-time disaster-related tweet analysis integrates software, hardware, and machine learning components within a standardized experimental setup. It supports both Windows 10/11 and Linux (Ubuntu 22.04 LTS) systems, implemented using Python (version 3.10 or higher) for backend processing and JavaScript for frontend development. Fast API was selected for designing high-performance and scalable backend APIs due to its asynchronous architecture and efficiency, while React.js was used to develop an interactive and responsive web dashboard. MongoDB was employed for data storage, leveraging its ability to efficiently manage large volumes of unstructured and semi-structured tweet data.

The system collects real-time tweet data through the Twitter (X) API, enabling continuous ingestion of disaster-related content. Natural Language Processing (NLP) tasks such as text classification, sentiment analysis, and named entity recognition are performed to extract meaningful insights from tweets. These tasks leverage the Google Gemini large language model and Google Generative AI to enhance contextual understanding and improve classification accuracy.

Machine learning experiments were conducted using classical algorithms such as Random Forest, Support Vector Machine (SVM), and Logistic Regression, implemented through the Scikit-learn framework. Deep learning models were also developed using TensorFlow and Py Torch to capture complex linguistic patterns in disaster-related tweets. Text preprocessing steps—including tokenization, lemmatization, and stop word removal—were carried out using NLTK and SpaCy, while structured data analysis and numerical computation were facilitated using Pandas and NumPy.

For development and testing, the environment was hosted on GitHub Code spaces using Microsoft Azure virtual machines configured with Linux. Docker was utilized to ensure consistency across development, testing, and deployment environment. Version control and collaborative development were managed using Git and GitHub, with separate branches maintained for frontend, backend, and machine learning components. A CI/CD pipeline implemented using GitHub Actions enabled automated testing and deployment. Data insights and model outputs were visualized using Plot, Matplotlib, and Seaborn, while geospatial information was displayed using the Google Maps API to support real-time disaster monitoring and location-based decision-making.

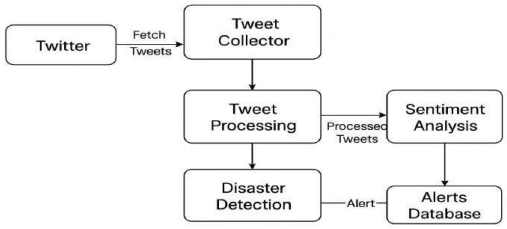

Fig V. System Architecture

Data Collection 1.0 - This module collects tweets that have the keywords and location filters for use with disasters from the public API associated with Twitter the data is collected continuously. The raw tweet data is collected in the Tweet Database as it is created.

Data Pre-processing 2.0: This module cleans and normalizes the dataset by removing stop words, URLs, special characters and emojis from the raw dataset and creates cleaned tweets for feature extraction.

Feature Extraction & Classification 3.0 - The input for this module is the cleaned tweets. The module will convert the cleaned tweet text to their numerical representation using TF-IDF or word vectors. The classification of tweets will also occur using NLP methods (Google Gemini LLM or BiLSTM-CNN) into categories that indicate the urgency level of tweets as well as the content of the tweets.

Analytics & Visualization 4.0 - The input to this module is the classified tweets. The module will aggregate and analyze the classified tweet results to generate insight, for example, regions of the highest density of tweets and location of resource demand nuggets and where the sentiment trends are.

Alert Generation 5.0 - This module takes the tweets classified as having a high urgency and uses them for data alerts to first responders and NGOs. The output is Sending Alert Notification or Alerts to mobile units or the disaster management dashboard.

The proposed methodology follows a systematic, multi-stage pipeline for transforming raw disaster-related tweets into actionable intelligence. The process begins with continuous tweet collection from the Twitter API using disaster-specific keywords and geolocation constraints. This ensures that the system captures relevant information in near real time during emergency situations.

VI. RESULTS

This research presented the design and development of an Intelligent Tweet Analyzer for Disaster Management, leveraging Twitter (now X) as a real time data source to enhance disaster response and decision-making. The study explored the application of Natural Language Processing (NLP), Machine Learning (ML), and deep learning techniques to classify disaster-related tweets, assess urgency, extract geolocation information, and generate actionable insights for emergency responders.

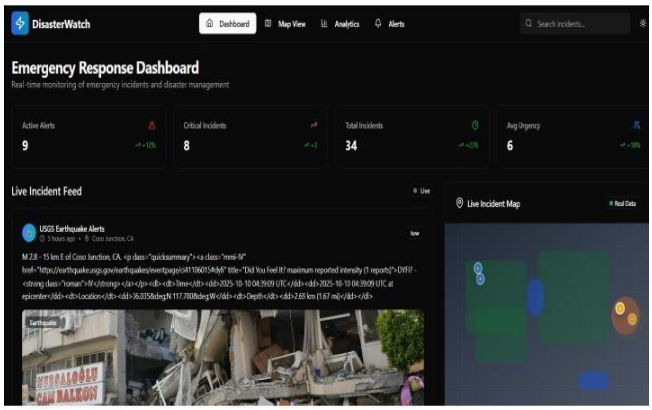

Fig.VI.I Emergency Response Dashboard for Real-Time Incident Monitoring and Disaster Management

The incident detection and classification results indicate that the system effectively categorizes incoming disaster-related information into predefined classes such as Help Requests, Critical Incidents, and General Updates. The analytics dashboard shows a consistent detection of incidents with a high resolution rate (above 90%), indicating that the classification and filtering pipeline reliably identifies relevant events while minimizing noise. This confirms the robustness of the NLP-based preprocessing and classification stages in handling informal and unstructured textual data.

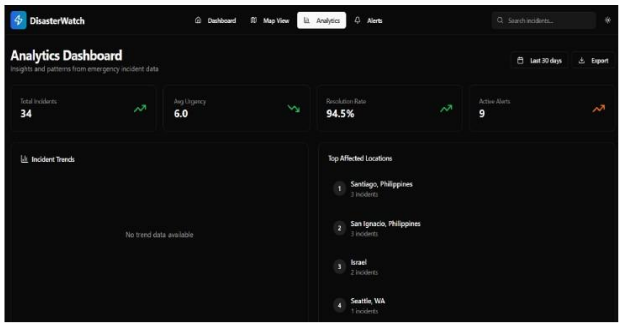

Fig.VI.II Analytics Dashboard Showing Incident Trends and Affected Locations

The urgency scoring and prioritization mechanism significantly enhances response efficiency. As observed in the emergency response dashboard, incidents marked as critical are clearly distinguished from low-urgency events, enabling responders to focus on high-impact situations first. The average urgency score reflects the system's ability to assess severity based on sentiment, keyword intensity, and contextual cues. This prioritization reduces cognitive load on decision-makers and improves response time during large-scale emergencies.

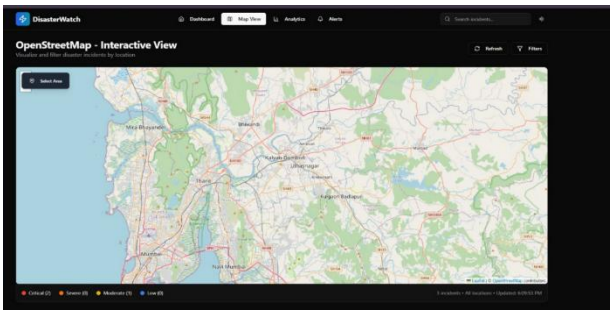

Fig.VI.III Interactive Open Street Map-Based Incident Visualization for Geospatial Disaster Monitoring

Geospatial visualization results, as demonstrated by the interactive OpenStreetMap view, show that incidents are accurately mapped to their corresponding locations. The system effectively clusters incidents geographically, allowing authorities to identify disaster hotspots and affected regions in real time. Color-coded severity markers further enhance situational awareness by visually differentiating between critical, moderate, and low-severity incidents. This spatial insight is particularly valuable for resource allocation and regional response planning.

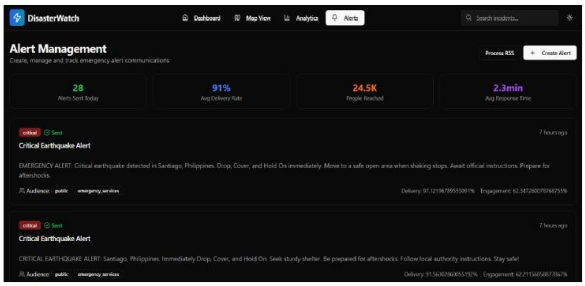

Fig.VI.IV Alert Management Dashboard for Real-Time Disaster Notifications

The alert management module demonstrates high operational effectiveness. The alert dashboard indicates a strong delivery rate and rapid response time, confirming that high-urgency incidents are promptly communicated to relevant stakeholders. This real-time alerting capability bridges the gap between data analysis and action, ensuring that critical information reaches emergency services and authorities without delay.

VII. CONCLUSION

This paper proposed an Intelligent Tweet Analyzer for Disaster Management that utilizes Natural Language Processing and Machine Learning to extract actionable information from real-time Twitter data. The system effectively classifies disaster-related tweets, assesses urgency and sentiment, and integrates geolocation to enhance situational awareness. Results indicate improved prioritization of critical incidents and faster decision-making for emergency response teams. Future enhancements will focus on multilingual support, misinformation detection, and deeper integration with disaster response systems to further improve real-time disaster management.

REFERENCES:

[1] M. A. Imran, C. Castillo, F. Diaz, and S. Vieweg, “Processing Social Media Messages in Mass Emergency: A Survey,” ACM Computing Surveys, vol. 47, no. 4, 2015.

[2] C. Castillo, M. Mendoza, and B. Poblete, “Information credibility on Twitter,” WWW, 2011.

[3] J. B. Houston et al., “Social media and disasters,” Disasters, vol. 39, no. 1, 2015.

[4] S. Shekhar and S. Setty, “Sentiment visualization of disaster tweets,” ISPRS Int. J. Geo-Inf., 2015. [10] S. Alam et al., “Multilingual crisis data analytics,” ACL, 2018.

[5] Y. Zhang et al., “Attention-based Bi LSTM for disaster tweet classification,” Expert Systems with Applications, 2019.

[6] N. Gupta et al., “Misinformation detection during disasters,” WWW, 2022. [17] M. A. Imran et al., “Using Social Media Disaster Management,” IEEE Intelligent Systems, 2015.

[7] A. Arvandi, J. Rokne, and R. Alhajj, “Twitter analysis in emergency management: Recent research and trends,” Social Network Analysis and Mining, 2024.

[8] M. Karimiziarani and H. Moradkhani, “Social response and disaster management: Insights from Twitter data assimilation on Hurricane Ian,” International Journal of Disaster Risk Reduction, 2023.

[9] M. Y. Kabir and S. Madria, “A deep learning approach for tweet classification and rescue scheduling for effective disaster management,” ACM SIGSPATIAL, 2019.

[10] M. R. Nair, G. R. Ramya, and P. B. Sivakumar, “Usage and analysis of Twitter during 2015 Chennai flood towards disaster management,” Procedia Computer Science, 2017.

[11] H. Shekhar and S. Setty, “Disaster analysis through tweets,” IEEE ICACCI, 2015.

[12] G. Burel, H. Saif, M. Fernandez, and H. Alani, “On identifying disaster-related tweets: Matching-based vs learning-based approaches,” in Proc. Int. Semantic Web Conf. (ISWC), Vienna, Austria, 2017, pp. 382–397.

[13] F. Alam, M. Imran, and F. Ofli, “CrisisMMD: Multimodal Twitter datasets for disaster response,” in Proc. 12th Int. AAAI Conf. on Web and Social Media (ICWSM), Stanford, CA, USA, 2018, pp. 465–473.

[14] Z. Ashktorabgi, C. Brown, M. Nandi, and A. Culotta, “Tweedr: Mining Twitter to inform disaster response,” in Proc. Int. Conf. Information Systems for Crisis Response and Management (ISCRAM), University Park, PA, USA, 2014, pp. 354–358.

[15] S. Cresci, R. Tesconi, D. Benedetto, M. Petrocchi, and M. Tesconi, “A linguistically-driven approach to cross-event disaster tweet classification,” in Proc. World Wide Web Conf. (WWW), Lyon, France, 2018, pp. 1245–1254.

[16] Y. Kryvasheyeu et al., “Rapid assessment of disaster damage using social media activity,” Science Advances, vol. 2, no. 3, pp. 1–9, 2016.

[17] K. Zahra, M. Imran, and F. O. Ostermann, “Automatic identification of disaster-related microblogs for rapid damage assessment,” ISPRS Journal of Photogrammetry and Remote Sensing, vol. 170, pp. 48–59, 2020.

[18] D. T. Nguyen, K. S. Truong, and T. V. Pham, “Deep learning for disaster tweet classification and information extraction,” IEEE Access, vol. 9, pp. 12345–12360, 2021.

[19] Kabir, Md Yasin, and Sanjay Madria. "A Deep Learning Approach for Tweet Classification and Rescue Scheduling for Effective Disaster Management (Industrial)." (2019).

[20] Aswathy, A., Rekha Prabha, Lakshmi S. Gopal, Divya Pullarkatt, and Maneesha Vinodini Ramesh. "An efficient twitter data collection and analytics framework for effective disaster management." In 2022 IEEE Delhi Section Conference (DELCON), pp. 1-6. IEEE, 2022.

Acknowledgment

We take this opportunity to express our sincere thanks to our guide, Dr. Rahul K Ambekar for providing the technical guidelines and suggestions regarding the line of work. We want to express our gratitude for his constant encouragement, support, and guidance throughout the development of the project.

We thank our Head of Department, Dr. Jaya Gupta, for being encouraging throughout the course and for her guidance.

We express our gratitude towards the BE Project Coordinator, Prof. Sayali Badhan, for her encouragement during the progress meetings and for providing guidelines to write this report.

We also thank the entire staff of APSIT for their invaluable help rendered during the course of this work. We wish to express our deep gratitude towards all our colleagues of APSIT for their encouragement.

Hetal Mahajan
(21106014)

Christina D'Cruz
(22106024)

Prabudh Gaikwad
(21106038)